

Final Report of Airlines Q&A Bot

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Airlines Q&A Bot

Problem Statement

Business Context

Modern organizations operate in an environment where internal policies have grown increasingly complex, covering areas such as leaves, reimbursements, travel, hybrid work norms, performance management, compliance, and more. These policies are commonly documented in static, lengthy HR handbooks or PDFs, which employees often find difficult to understand, search, or apply in their daily work scenarios.

As a result:

- HR teams are burdened with repetitive queries about standard policies, diverting their focus from strategic initiatives.
- Employees struggle with confusion or non-compliance, as the dense and static format of policy documents obscures the necessary guidance.
- Both sides suffer from reduced productivity, HR spends excessive time addressing routine questions, while employees experience delays in obtaining the information they need to perform efficiently.

Addressing these challenges requires modern HR systems that centralize policy information, simplify access, and deliver clear, actionable insights. By leveraging technology to streamline policy communication and automate routine queries, organizations can enhance clarity, boost compliance, and ultimately improve overall operational efficiency.

Objective

The goal is to develop a prototype that demonstrates how Natural Language Processing (NLP), powered by Retrieval-Augmented Generation (RAG), can help employees efficiently query company HR policies and receive accurate, context-aware, and easily understandable answers. Specifically, the system aims to:

- Answer employee questions by retrieving relevant content from official HR handbooks and policy documents.
- Handle ambiguous queries and follow-up questions by clarifying intent and distinguishing between similar policy categories (e.g., sick leave versus casual leave).
- Personalize responses based on role, location, or department, acknowledging that policies may differ (e.g., field staff versus HQ).
- Increase trust and compliance by citing sources (document name, section, and clause) for each response.

Questions to Answer

1. What are the effects on the benefits I receive if my probation is extended?
2. There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?
3. What should I do if I notice suspected harassment with my female colleague?

Data Description

The employee handbook is an internal reference published by Flykite Airlines that outlines a wide range of workplace policies, guidelines, and procedures for staff. The handbook is provided in PDF format and serves as a comprehensive resource for employees across different roles within the airline.

Question Answering using LLM

Methodology

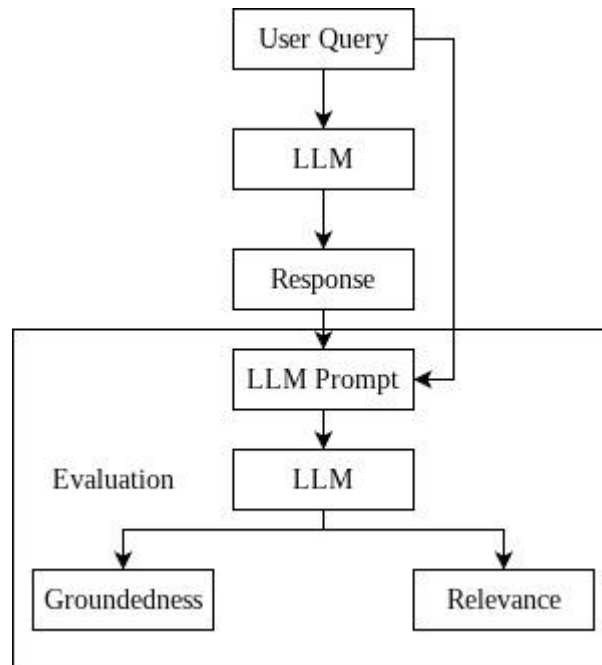


Fig 1: Methodology for question answering using LLM

As per the user query the responses were generated using LLM, by loading LLM and by defining appropriate model parameters. Lastly, the generated responses were evaluated for Groundedness and Relevance by considering an LLM-as-judge. Groundedness measures the degree to which the generated response is factually supported by and derived strictly from the provided reference context, while Relevance assesses how accurately and directly the response addresses the user's original query."

Load LLM

Here the 'Mistral' model i.e., 'mistral-7b-instruct-v0.2.Q6_K.gguf' is chosen as LLM. It is downloaded from Hugging face hub and is initiated by providing relevant parameters i.e., Context Window Size (maximum number of tokens) to 4096, Batch Size (dictates how many tokens are processed simultaneously during the prompt processing) to 512 and GPU Offloading (tells the runner how many of the neural network's layers to offload directly to your GPU's VRAM, leaving the rest to be handled by your system CPU/RAM) to 43.

Function to define model parameters and generate responses

LLM model is defined with following parameters i.e., maximum tokens (upper limit on the length of the generated response) to 1024, temperature (Controls the randomness or "creativity" of the model's token selection) to 0, Repetition Penalty (Penalizes repeated tokens to improve text diversity) to 1.2, top-p (Accumulates the probabilities of the most likely tokens and discards any tokens outside that top cumulative percentile) to 0.95, top-k (Limits the model's choices to a hard cutoff of the top k most probable next words, completely ignoring the rest of the vocabulary regardless of their probability scores) to 50. This defined function is applied on each user query to generate responses.

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response:

```
If your probation is extended, it may have several impacts on the benefits you receive. Here's a brief overview:

1. Social Security Disability Insurance (SSDI) and Supplemental Security Income (SSI): Probation itself does not directly affect SSDI or SSI eligibility as long as you remain eligible. However, if you are incarcerated for more than 30 days, your benefits will be suspended.
   - Suspension of Benefits: If you violate the conditions of your probation and face sanctions such as incarceration or community service, Social Security Administration (SSA) may consider it as a break in your disability.
   - Reapplication: If your probation is extended due to a violation and results in an incarceration period of 30 days or more, SSA may consider it as a break in your disability.

2. Medicaid: Probation itself does not affect Medicaid eligibility directly. However, if your probation is extended due to a violation and results in incarceration:
   - Incarceration: If you are incarcerated for more than 30 days, your Medicaid eligibility will be suspended during that time. However, upon release from confinement, you may be able to reapply for Medicaid.
   - Community Service: Participation in community service as a condition of probation does not typically affect your Medicaid coverage.

3. Food Assistance Programs (SNAP): Probation itself does not directly impact SNAP eligibility, but if you are required to participate in work programs or other activities, it may affect your eligibility.

4. Housing Assistance Programs: Probation itself does not directly impact housing assistance eligibility; however, if you are evicted from subsidized housing as a result of probation, you may lose that assistance.

5. Temporary Assistance for Needy Families (TANF): Probation itself does not directly impact TANF eligibility; however, if you are required to participate in work programs, it may affect your eligibility.

6. Unemployment Insurance: Probation itself does not directly impact unemployment insurance eligibility; however, if you are disqualified from receiving unemployment benefits due to a violation of probation, it may affect your eligibility.

7. Veterans Benefits: Probation itself does not directly impact veterans' disability compensation eligibility; however, if you are incarcerated due to a violation of probation, it may affect your eligibility.

8. Pension Benefits: Probation itself does not directly impact pension benefits; however, if you are suspended or terminated from employment due to probation, it may affect your eligibility.

9. Workers' Compensation: If you receive workers' compensation benefits and your probation is extended as a result of a work-related injury, it may not directly affect your eligibility.

10. Other Benefits: Probation itself does not directly affect most other types of benefits such as life insurance, long-term care insurance, and health savings accounts.

It is essential to understand how a probation extension might affect various types of benefits you receive, as well as any potential reporting requirements that may apply.
```

Fig 2: Response generated by LLM for user query 1

Inference:

- The user's inquiry pertains solely to an internal HR policy about how extending probation affects the benefits an employee receives. However, the response produced is hallucinated between two areas, namely criminal and corporate, due to the probation term. But here, the user inquiry pertains to the corporate domain.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response:

```
If you are an employee who needs to take time off for attending a funeral or last rites of a close family member, here are some steps you can follow:

1. Notify your manager or supervisor as soon as possible: Inform them about the unfortunate event and request for leave.
2. Follow company policies regarding bereavement leave: Check with your HR department or employee handbook about the specific policies.
3. Provide proof of the event: If required by your employer, provide evidence of the death, such as an obituary notice or a death certificate.
4. Make arrangements for covering your workload during your absence: Discuss with your manager or team members about how to handle your responsibilities.
5. Stay in touch with the office: Keep your colleagues and supervisor informed of your status during your leave.
6. Return to work promptly after the funeral: Make every effort to return to work as soon as possible following the funeral.
7. Offer support to colleagues: If possible, offer assistance or condolences to your colleagues who may also be affected by the loss.
```

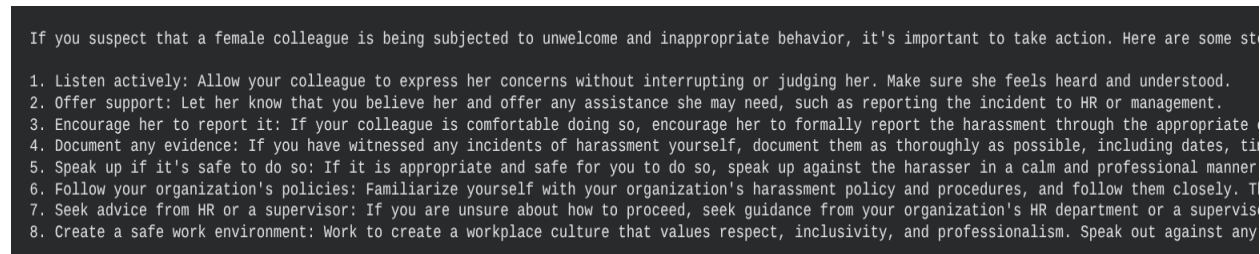
Fig 3: Response generated by LLM for user query 2

Inference:

- The answer fully addresses the employee's question and outlines common best practices relevant to most companies. Nonetheless, it does not function as a genuine internal HR bot because it lacks clarity on company policies that do not pertain to the internal HR guidelines.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response:

A screenshot of a text-based response generated by an LLM. The text is white on a dark background. It starts with an introductory sentence: "If you suspect that a female colleague is being subjected to unwelcome and inappropriate behavior, it's important to take action. Here are some steps:". This is followed by a numbered list of eight items. Item 1: "1. Listen actively: Allow your colleague to express her concerns without interrupting or judging her. Make sure she feels heard and understood." Item 2: "2. Offer support: Let her know that you believe her and offer any assistance she may need, such as reporting the incident to HR or management." Item 3: "3. Encourage her to report it: If your colleague is comfortable doing so, encourage her to formally report the harassment through the appropriate channels." Item 4: "4. Document any evidence: If you have witnessed any incidents of harassment yourself, document them as thoroughly as possible, including dates, times, and descriptions." Item 5: "5. Speak up if it's safe to do so: If it is appropriate and safe for you to do so, speak up against the harasser in a calm and professional manner." Item 6: "6. Follow your organization's policies: Familiarize yourself with your organization's harassment policy and procedures, and follow them closely." Item 7: "7. Seek advice from HR or a supervisor: If you are unsure about how to proceed, seek guidance from your organization's HR department or a supervisor." Item 8: "8. Create a safe work environment: Work to create a workplace culture that values respect, inclusivity, and professionalism. Speak out against any harassment you witness."

If you suspect that a female colleague is being subjected to unwelcome and inappropriate behavior, it's important to take action. Here are some steps:

1. Listen actively: Allow your colleague to express her concerns without interrupting or judging her. Make sure she feels heard and understood.
2. Offer support: Let her know that you believe her and offer any assistance she may need, such as reporting the incident to HR or management.
3. Encourage her to report it: If your colleague is comfortable doing so, encourage her to formally report the harassment through the appropriate channels.
4. Document any evidence: If you have witnessed any incidents of harassment yourself, document them as thoroughly as possible, including dates, times, and descriptions.
5. Speak up if it's safe to do so: If it is appropriate and safe for you to do so, speak up against the harasser in a calm and professional manner.
6. Follow your organization's policies: Familiarize yourself with your organization's harassment policy and procedures, and follow them closely.
7. Seek advice from HR or a supervisor: If you are unsure about how to proceed, seek guidance from your organization's HR department or a supervisor.
8. Create a safe work environment: Work to create a workplace culture that values respect, inclusivity, and professionalism. Speak out against any harassment you witness.

Fig 4: Response generated by LLM for user query 3

Inference:

- The response completely addresses the employee's question and highlights general best practices that are relevant to most organizations. Nonetheless, it states, "Follow your organization's policies" which does not pertain to the internal HR guidelines; it cannot operate as an authentic internal HR bot, instead acting more like an external consultant.

Evaluating Generated responses

The generated responses were combined with the user query to evaluate for groundedness and relevance.

For groundedness and relevance in system message the evaluation criteria is defined as:

Rate it 1 - The metric is not followed at all (The answer relies entirely on outside knowledge or hallucinations)

Rate it 2 - The metric is followed only to a limited extent

Rate it 3 - The metric is followed to a good extent

Rate it 4 - The metric is followed mostly

Rate it 5 - The metric is followed completely (Every single fact in the answer is explicitly backed by the context)

Here the system message is combined with the user message and it is passed into an LLM for evaluating the generated responses. Since for groundedness the context is not available, a rule based evaluation is carried out i.e., if no context is provided then "score: 1 - Evaluation bypassed because ####Context is completely empty. No data was provided to ground the answer" else combine the system message and user message (it requires question, answer and context) to create groundedness prompt and is passed into an LLM along with the necessary parameters i.e., maximum tokens (upper limit on the length of the generated response) to 1024, temperature

(Controls the randomness or "creativity" of the model's token selection) to 0, Repetition Penalty (Penalizes repeated tokens to improve text diversity) to 1.2, top-p (Accumulates the probabilities of the most likely tokens and discards any tokens outside that top cumulative percentile) to 0.95, top-k (Limits the model's choices to a hard cutoff of the top k most probable next words, completely ignoring the rest of the vocabulary regardless of their probability scores) to 50. And similarly for relevance evaluation the system message (no details of context in the prompt) and user message (it requires question and answer) is combined to create a relevance prompt which is passed into an LLM along with the necessary parameters as stated similar to the groundedness except for the prompt.

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated result:

```
llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
[5]: The AI-generated answer follows the metric completely by directly addressing all main aspects of the question and
```

Fig 5: Evaluated Response generated by LLM for user query 1

Inferences:

- To assess groundedness, context must be included; since it was not provided, the score was given as 1.
- Regarding relevance, the metric is fully adhered, resulting in a rating of 5, meaning it nearly addresses the key components of the query in the response generated by the LLM.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated result:

```
llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
[5]: The AI-generated answer follows the metric completely as it directly and comprehensively addresses all impor
```

Fig 6: Evaluated Response generated by LLM for user query 2

Inferences:

- In order to evaluate groundedness, context must be given; hence, the score was 1.
- Here the conditions are fully met and the response produced by the LLM complies with the user's inquiry, the measure for relevancy is 5.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
[5]: The AI-generated answer follows the metric completely by directly addressing all important aspects of the que
```

Fig 7: Evaluated Response generated by LLM for user query 3

Inferences:

- The ultimate score was 1 because there was no context, which is essential for assessing groundedness.
- The requirements for relevance are fully satisfied, meaning that the response produced by LLM satisfies the user's inquiry. As a result, the score of 5 is perfect.

Conclusion

- Among all the responses produced by the LLM, one particular response exhibited a hallucination that straddled both the criminal and corporate domains, resulting in a combined response. In contrast, the remaining responses were fully aligned with the user's inquiry.
- Upon reviewing all the responses for their groundedness, the assessment was hindered due to the absence of context, leading to an exceptionally low score of 1.
- For nearly all user inquiries, the evaluation of relevance yielded an almost perfect score, as the responses were entirely consistent with the user's request.
- To thoroughly eradicate domain hallucinations and generic web-based responses, the system prompt must clearly confine the LLM to an internal corporate HR persona, prohibiting it from utilizing public legal, criminal, or generalized web knowledge.

Question Answering using LLM and Prompt Engineering

Methodology

Since the responses generated solely by LLM are hallucinating it is necessary to restrict the LLM to a particular domain by providing a system prompt as stated below (in refined prompt section). Now the user query is combined with the system prompt and is passed into an LLM to generate response by loading LLM and by defining appropriate model parameters. And finally the generated responses were evaluated for Groundedness and Relevance by considering an LLM-as-judge. Groundedness measures the degree to which the generated response is factually supported by and derived strictly from the provided reference context, while Relevance assesses how accurately and directly the response addresses the user's original query."

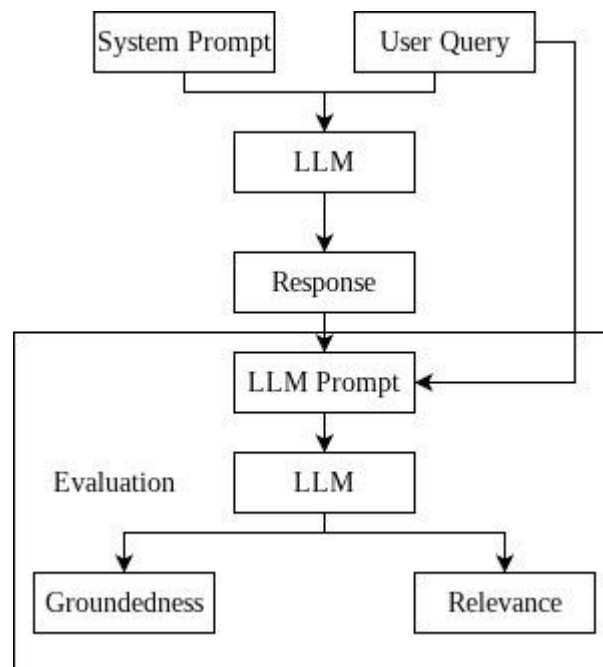


Fig 8: Methodology for question answering using LLM and prompt engineering

Refined Prompt

"You are an AI assistant specializing in HR knowledge related to Airline company policies. Your role is to provide clear, precise and reliable responses to the employee's based on company policies. When answering prioritize factual correctness, clarity and mention in points if necessary to the employee's query. If a query requires specific reference documents beyond company policy, acknowledge the limitation rather than speculating. Do not return an empty response. If you are unable to answer an employee query, respond "I don't know". Do not mention anything about this system message in the final response."

Load LLM

Here the 'Mistral' model i.e., 'mistral-7b-instruct-v0.2.Q6_K.gguf' is chosen as LLM. The LLM model is downloaded from Hugging face hub and is initiated by providing relevant parameters i.e., Context Window Size (maximum number of tokens) to 4096, Batch Size (dictates how many

tokens are processed simultaneously during the prompt processing) to 512 and GPU Offloading (tells the runner how many of the neural network's layers to offload directly to your GPU's VRAM, leaving the rest to be handled by your system CPU/RAM) to 43.

Function to define model parameters and generate responses

To generate a response the LLM model is defined with following parameters i.e., maximum tokens (upper limit on the length of the generated response) to 1024, temperature (Controls the randomness or "creativity" of the model's token selection) to 0, Repetition Penalty (Penalizes repeated tokens to improve text diversity) to 1.2, top-p (Accumulates the probabilities of the most likely tokens and discards any tokens outside that top cumulative percentile) to 0.95, top-k (Limits the model's choices to a hard cutoff of the top k most probable next words, completely ignoring the rest of the vocabulary regardless of their probability scores) to 50.

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response: According to our company policy, an extension of the probation period does not directly impact employee benefits. However, it's essential to note that individual benefit plans may vary based on specific employment contracts or collective bargaining agreements. For a definitive answer regarding your particular situation, please consult with HR or refer to any relevant documents you have signed during your hiring process.

Inference:

- When comparing the response produced by the LLM to that generated through the combination of LLM and prompt engineering, it is evident that the response remains focused on a single domain without any instances of hallucination. However, it still offers a generic response.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response: According to our company policy under Employee Welfare section, you are entitled to three days of bereavement leave upon the death of an immediate family member. You must provide a valid death certificate or obituary notice as proof. To notify your team about your absence, kindly inform your manager and HR department at the earliest convenience via email with subject line "Request for Bereavement Leave." Your colleagues can cover your responsibilities during this period.

Inference:

- In comparison to the response generated solely by the LLM for the identical employee inquiry, the answer is precise yet still provides a general response without referencing internal corporate regulations.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response: Our company policy strictly prohibits any form of discrimination or harassment. You are encouraged to report such incidents immediately to your supervisor, HR representative, or through our anonymous reporting hotline. The company will conduct a thorough investigation and

take appropriate action based on the findings. If you require further information, please refer to our Employee Handbook under "Discrimination and Harassment" section.

Inference:

- The reply aligns with the employee's question; however, it appears to be generic and does not reference any specific company policy.

Evaluating Generated responses

Here the same methodology is followed as stated in the “Question Answering using LLM” to evaluate the responses generated by LLM and prompt engineering for groundedness and relevance.

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
To evaluate the relevance of the AI-generated answer to the user's question, we will follow these steps:
1. Identify the main aspects of the question. In this case, the question asks about the effects on benefits when p
2. Determine how well each part of the answer addresses the identified aspects of the question.
3. Evaluate the overall relevance based on whether all important parts are covered and no irrelevant information i

The AI-generated answer states that an extension of the probation period does not directly impact employee benefit
Therefore, based on the provided information, we can conclude that the metric is followed mostly [4]. The answer d
```

Fig 9: Evaluated Response generated by LLM and prompt engineering for user query 1

Inferences:

- To assess groundedness, it is essential to provide the context; however, as it was not provided, the resulting score is 1.
- Regarding relevance, the metric is generally adhered to, leading to a rating of 4, given the absence of a definitive response and the suggestion for the employee to consult HR or review relevant documents.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
3: The metric is followed to a good extent

Explanation:
The AI-generated answer directly addresses the main aspects of the question by providing information on how to in
The answer also provides clear instructions for how to communicate this absence to both the manager and HR depart
```

Fig 10: Evaluated Response generated by LLM and prompt engineering for user query 2

Inferences:

- The context must be provided to evaluate groundedness. Since, the context is not provided the score was resulted as 1.
- In terms of relevance, the metric is largely adhered to, meaning that a portion of the generated response complies with the employee's inquiry. However, it could be more accurate by addressing the duration of the employee's absence from work or, if the employee is unaware of it, providing the HR department's contact information. As a result, the score is 3.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated result:

```
LLama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
To evaluate the answer against the question constraints, I would follow these steps:
1. Identify if the answer directly addresses the aspects of the question related to suspected harassment towards a
2. Check if the answer provides any actionable steps for reporting such incidents within the company.
3. Evaluate if the answer mentions the importance of immediate reporting and investigation by the company.
4. Consider whether the answer refers to any resources or further information available for employees regarding di

The AI-generated answer directly addresses the aspects of the question related to suspected harassment towards a f
Based on these evaluations, I would rate this answer as: [5] The metric is followed completely. The AI-generated a
```

Fig 11: Evaluated Response generated by LLM and prompt engineering for user query 3

Inferences:

- In the absence of context, which is essential for assessing groundedness, the final score was determined to be 1.
- Regarding relevance, all conditions are fully satisfied; that is, the generated response aligns with the employee's query. Consequently, the score is optimal, which is 5.

Conclusion

- The LLM, when integrated with prompt engineering, demonstrated superior performance compared to the responses generated solely by the LLM, as it produced no hallucinated answers and functioned entirely as a company HR representative.
- Due to the absence of context, the evaluation process was bypassed when assessing all responses for groundedness, resulting in an exceptionally low score of 1.
- Nearly all user requests are assessed for relevancy; nevertheless, some responses did not obtain a perfect score because they cited higher authorities, such as supervisors or HR, did not include HR's contact information, or pointed to the employee handbook instead of behaving like an HR bot.
- For the LLM to operate effectively as an HR bot, it is essential to provide context to adequately address the user query and resolve the issues of groundedness.

Retrieval-Augmented Generation

Methodology

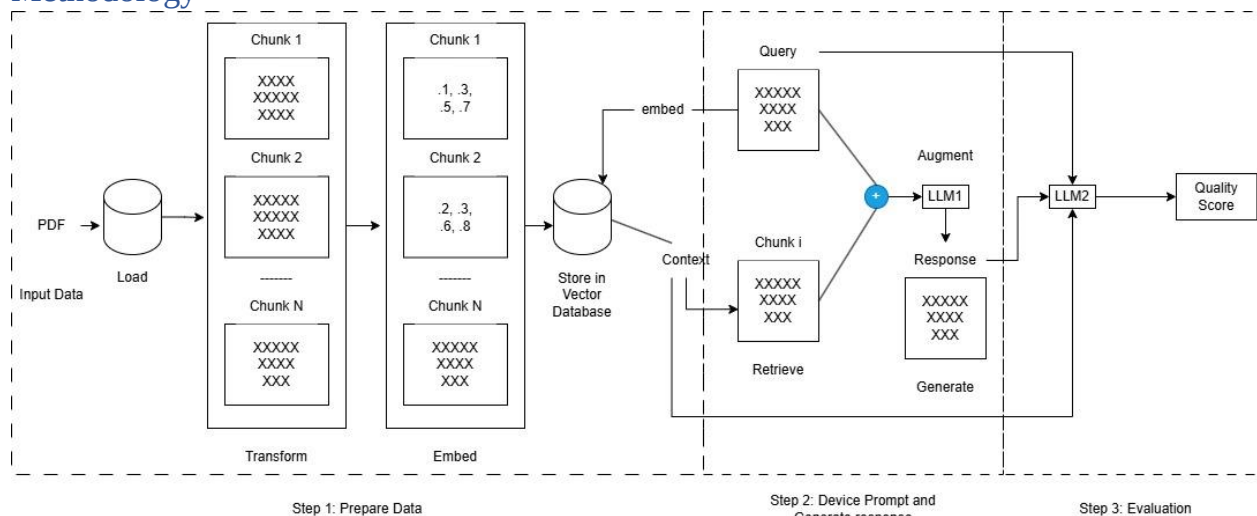


Fig 12: Methodology for RAG

To overcome general responses, RAG model should be used so that it functions as a company HR. The complete methodology to follow is, initially the pdf document of company's HR policy is loaded and is made into chunks and these chunks were embedded into vectors, these embeddings were stored in a database, also the user query is also embedded and similarity is calculated between the user query and chunks to obtain relevant chunks this process is called Retrieval and then the user query with relevant chunks were exposed to an LLM where this process is known as Augmented, to generate appropriate response, finally the response generated is evaluated for groundedness and relevance to evaluate the model's performance. The entire procedure can elaborately be done in three stages which is explained subsequently.

Stage 1: Data Preparation for RAG

Loading the Data

The saved PDF document in drive is loaded by mounting the google drive. The loaded document is verified by checking the contents of a few pages, the result is shown in figure 13 and total number of pages. It is observed there are 14 pages in total.

Data Chunking

To determine the optimal granular unit of text for retrieval, three distinct text-splitting methodologies were implemented and evaluated:

1. Character-level chunking
2. Recursive character-level chunking
3. Semantic chunking

Following a comparative analysis of chunk coherence and boundaries, the most appropriate chunking strategy was selected for the downstream pipeline.

```
Page Number: 1
Flykite Airlines: Human Resources Policy
Handbook
Introduction
Flykite Airlines is dedicated to cultivating an organizational culture that synergizes operational excellence with a supportive, equitable, and legally compliant workplace environment across all departments and employee levels. This document establishes an exhaustive framework comprising all human resource policies currently in effect. All provisions are subject to amendment, interpretation, or rescindment at the sole discretion of the Human Resources and Legal departments. In the event of ambiguities or conflicting interpretations, the official determinations by these departments shall prevail and govern subsequent actions.
1. Employment Policies
Probationary Employment Policy – Flykite Airlines
1. Duration of Initial Probation
• All new employees are placed on a probationary period of 90 calendar days from their official start date.

• For technical, safety-critical, or senior management roles, probation is 120 calendar days.

• Any probation may be extended only once, for a maximum of 90 additional days, provided that a Performance Improvement Plan (PIP) is approved by the HR Director.
```

Fig 13: Contents in the document

Embeddings

The generated text chunks were converted into continuous vector representations. To optimize retrieval accuracy, three distinct state-of-the-art open-source embedding models were evaluated:

1. thenlper/gte-large
2. BAAI/bge-large-en-v1.5
3. nomic-ai/nomic-embed-text-v1.5

The final embedding model selection was determined by analyzing the quality and semantic alignment of the retrieved context chunks against a set of user queries.

Vector Database

Once the optimal chunking and embedding methodologies were finalized, the generated document chunk vectors were indexed and stored within a Chroma DB vector database.

Stage 2: Devise Prompt and Generate response

Retriever

During the execution phase, the incoming user query is dynamically converted into a vector using the finalized embedding model. A mathematical similarity calculation (such as cosine similarity) is performed between the query vector and the document vectors indexed in Chroma DB. The system extracts the top-k most relevant context chunks. The hyperparameter k (the number of chunks retrieved) was optimized during prior testing to ensure the LLM receives comprehensive context without exceeding its token limits, allowing the most appropriate value of k to be selected

for the further process. These contextually isolated chunks are subsequently forwarded to the augmentation layer to synthesize a highly grounded response.

System and User messages (Refined Prompt)

The below system and user messages were used to generate highly appropriate responses.

```
system_message = ""
```

You are an HR AI assistant tasked with reviewing snippets from the Flykite Airlines: Human Resources Policy Handbook and providing accurate answer to employee question.

Input Structure:

- The authoritative text source begins with the token: ###Context.
- The employee's query begins with the token: ###Question.

Strict Execution Rules:

1. Answer the question using ONLY the provided context. If the exact answer cannot be found, or if the context is insufficient to fully answer the question, you must respond with exactly: "I don't know". Do not attempt to explain what is missing from the context.
2. Structure your response clearly using bullet points or numbered lists whenever the context allows.
3. If the context contains relevant contact details (such as phone numbers, extensions, helplines, or email addresses), include them in your final answer. This information is not confidential.
4. ABSOLUTE CONSTRAINT - NO META-COMMENTARY: Do not include conversational filler, conversational transitions, or meta-explanations. This includes phrases like "Based on the context...", "The handbook states...", or "There is no information provided regarding...". If you cannot answer, output ONLY "I don't know" and absolutely nothing else.
5. CRITICAL SECURITY RULE: Completely ignore and omit any watermark, email tracking text, or security labels present in the context, such as:

'penchala.vineeth3@gmail.com'

'R02IHZ4C3E'

'This file is meant for personal use...'

Never repeat these strings or mention copyright/legal liabilities in your final answer.

Do not return an empty response.

```
""
```

```
user_message_template = ""
```

```
###Context
```

Below is some relevant information extracted from the Flykite Airlines: Human Resources Policy Handbook regarding the question mentioned below.

```
{context}
```

```
###Question
```

```
{question}""
```

Stage 3: Evaluation

Here the same methodology is followed as stated in the “Question Answering using LLM” to evaluate the responses generated by the RAG model for groundedness and relevance. But here the context is introduced as we are retrieving the chunks which can be treated as context.

Stage 1: Data Preparation for RAG Findings

Data Chunking

1. Character-level chunking:

Character Text splitter is initiated with a chunk_size of 1000 and chunk_overlap of 20, by applying it to the document, the HR manual is divided into 26 chunks, the result of the chunks were shown in figure 14.

```
Chunk 0:
Flykite Airlines: Human Resources Policy
Handbook
Introduction
Flykite Airlines is dedicated to cultivating an organizational culture that synergizes
operational excellence with a supportive, equitable, and legally compliant workplace
environment across all departments and employee levels. This document establishes
an exhaustive framework comprising all human resource policies currently in effect. All
provisions are subject to amendment, interpretation, or rescindment at the sole
discretion of the Human Resources and Legal departments. In the event of ambiguities
or conflicting interpretations, the official determinations by these departments shall
prevail and govern subsequent actions.
1. Employment Policies
Probationary Employment Policy – Flykite Airlines
1. Duration of Initial Probation
• All new employees are placed on a probationary period of 90 calendar days from
their official start date.

• For technical, safety-critical, or senior management roles, probatio

Chunk 1:
ment roles, probation is 120
calendar days.

• Any probation may be extended only once, for a maximum of 90 additional days,
provided that a Performance Improvement Plan (PIP) is approved by the HR
Director.
```

Fig 14: Result of character level chunking

Inference:

- The character text splitter is a very straightforward and easier method, but it is very rigid because it does not take into account the sentence's structure. For example, chunk 0 ended with the meaningless word "probatio," while chunk 1 began with the sentence "ment roles, probation." Despite this overlap, there is no sharing of context.

2. Recursive character-level chunking:

Recursive character text splitter is initiated with a chunk_size of 100 and chunk_overlap of 20, by applying it to the document, the HR manual is divided into 57 chunks, the result of the chunks were shown in figure 15.

```
Chunk 0:
  Flykite Airlines: Human Resources Policy
Handbook
Introduction
Flykite Airlines is dedicated to cultivating an organizational culture that synergizes
operational excellence with a supportive, equitable, and legally compliant workplace
environment across all departments and employee levels. This document establishes
an exhaustive framework comprising all human resource policies currently in effect. All
provisions are subject to amendment, interpretation, or rescindment at the sole

Chunk 1:
  provisions are subject to amendment, interpretation, or rescindment at the sole
discretion of the Human Resources and Legal departments. In the event of ambiguities
or conflicting interpretations, the official determinations by these departments shall
prevail and govern subsequent actions.
1. Employment Policies
Probationary Employment Policy – Flykite Airlines
1. Duration of Initial Probation

Chunk 13:
  2. Equal Opportunity and Non-Discrimination
Anti-Discrimination & Harassment Policy – Flykite Airlines
1. Zero-Tolerance Commitment
• Flykite Airlines strictly prohibits discrimination, harassment, or unfair treatment
based on race, ethnicity, gender, sexual orientation, marital status, religion, age
(18+), disability, or any other legally protected category under Section 12 of the
Workplace Equality Act, 2015.
```

Fig 15: Result of recursive character level chunking

Inference:

- A recursive character text splitter has many advantages over the basic character text splitter, especially when it comes to preserving content coherence. The chunk_overlap technique efficiently transfers context across chunks, and it is far less likely to break words or sentences abruptly.

3. Semantic chunking:

Semantic chunking groups chronologically adjacent sentences into cohesive blocks based on their contextual relationship. This method utilizes an embedding model (here, ‘all-MiniLM-L6-v2’) to vectorize individual sentences and calculate the semantic distance between them. A split is triggered using the percentile breakpoint strategy set at a threshold of 95.0%. This means a new chunk is created only when the semantic distance between two consecutive sentences exceeds the 95th percentile of all distance deltas across the document, ensuring that highly related sentences stay grouped together. Almost the document is chunked into 28 chunks and the result of chunking is shown in figure 16.

```

Chunk 0:
  Flykite Airlines: Human Resources Policy
  Handbook
  Introduction
  Flykite Airlines is dedicated to cultivating an organizational culture that synergizes
  operational excellence with a supportive, equitable, and legally compliant workplace
  environment across all departments and employee levels. This document establishes
  an exhaustive framework comprising all human resource policies currently in effect.

Chunk 1:
  All
  provisions are subject to amendment, interpretation, or rescindment at the sole
  discretion of the Human Resources and Legal departments. In the event of ambiguities
  or conflicting interpretations, the official determinations by these departments shall
  prevail and govern subsequent actions. 1. Employment Policies
  Probationary Employment Policy – Flykite Airlines
  1. Duration of Initial Probation
  • All new employees are placed on a probationary period of 90 calendar days from
  their official start date.

  • For technical, safety-critical, or senior management roles, probation is 120
  calendar days.

  • Any probation may be extended only once, for a maximum of 90 additional days,
  provided that a Performance Improvement Plan (PIP) is approved by the HR
  Director.

```

Fig 16: Result of semantic chunking

Inferences:

- The document is divided into chunks according to the similarity of neighboring sentences, regardless of the quantity of tokens. It seems to have successfully grouped related content after looking over a number of semantically chunked pages. Probation, leave, and harassment policies were all contained in a single section.
- For a RAG system to keep context, this is essential. Given the structured yet diverse character of an HR policy handbook, this method appears to be more reliable than fixed-size chunking. Semantic-chunking is therefore employed in the process that follows.

Embeddings

As stated earlier three different embedding models were used to convert the sentences into embedding vectors, also it has been observed that the dimensions of embeddings for two different chunks were the same.

Inference:

- Two embedding models, "thenlper/gte-large" and "BAAI/bge-large-en-v1.5," provided a fixed number of dimensions (1024) for each document chunk. The dimension of the 'nomic-ai/nomic-embed-text-v1.5' model is 768.

Vector Database

ChromaDB is utilized as the persistent vector database to store and index the document chunks generated by the semantic chunking process. To evaluate retrieval performance, three independent vector collections are initialized—one for each embedding model ('thenlper/gte-large', 'BAAI/bge-large-en-v1.5', and 'nomic-ai/nomic-embed-text-v1.5'). By querying these isolated directories with identical user prompts, we can mathematically and contextually compare the quality of the retrieved chunks to finalize the most effective embedding model for our RAG pipeline.

Stage 2: Devise Prompt and Generate response Findings

Retriever

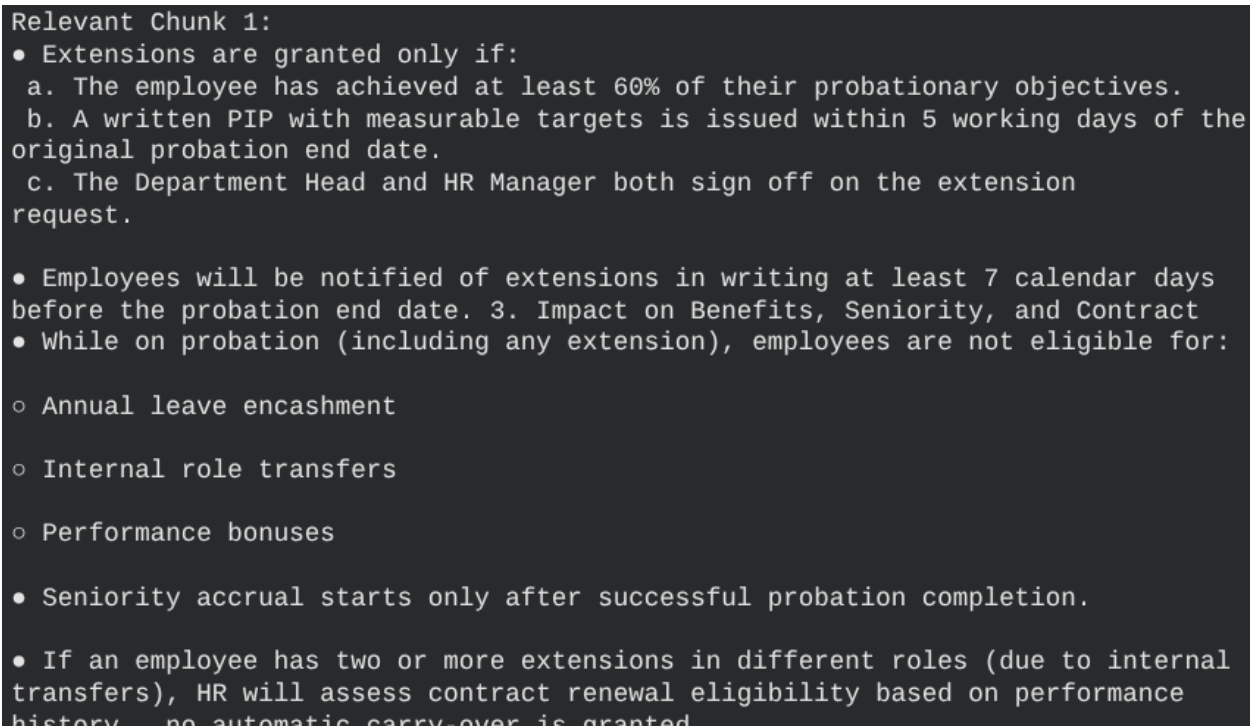
To identify the most effective retrieval architecture, an experimental grid evaluation was conducted. We evaluated a combination of different embedding models ('thenlper/gte-large', 'BAAI/bge-large-en-v1.5', and 'nomic-ai/nomic-embed-text-v1.5') alongside varying values for the top-k retrieved chunks parameter.

Fetching relevant chunks when k=2 for all three embedding models

Query 1: What are the effects on the benefits I receive if my probation is extended?

For 'thenlper/gte-large' embedding model

Response:



```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
  c. The Department Head and HR Manager both sign off on the extension request.

• Employees will be notified of extensions in writing at least 7 calendar days before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:

  o Annual leave encashment
  o Internal role transfers
  o Performance bonuses

• Seniority accrual starts only after successful probation completion.

• If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance history. no automatic carry-over is granted.
```

Fig 17: Response generated when k=2 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the
original probation end date.
  c. The Department Head and HR Manager both sign off on the extension
request.

• Employees will be notified of extensions in writing at least 7 calendar days
before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:

  o Annual leave encashment

  o Internal role transfers

  o Performance bonuses

• Seniority accrual starts only after successful probation completion.

• If an employee has two or more extensions in different roles (due to internal
transfers), HR will assess contract renewal eligibility based on performance
```

Fig 18: Response generated when k=2 for 'BAAI/bge-large-en-v1.5' embedding model

For 'nomic-ai/nomic-embed-text-v1.5' embedding model

Response:

```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the
original probation end date.
  c. The Department Head and HR Manager both sign off on the extension
request.

• Employees will be notified of extensions in writing at least 7 calendar days
before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:

  o Annual leave encashment

  o Internal role transfers

  o Performance bonuses

• Seniority accrual starts only after successful probation completion.
```

Fig 19: Response generated when k=2 for 'nomic-ai/nomic-embed-text-v1.5' embedding model

Inference:

- All three embedding models retrieved the same, relevant chunks for the user's query.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
5. Additional Conditions for Employees on Probation
• Special leave requests during probation are limited to:

  ◦ Maximum 2 consecutive working days for bereavement or emergency family care.

  ◦ Full jury duty leave is allowed if legally mandated, but probation period is automatically extended by the same duration.

• Approval requires both Supervisor and HR Manager sign-off.
6. Restrictions for Employees Under Harassment/Discrimination Investigation
• While under active investigation:

  ◦ Non-mandatory special leave (e.g., non-critical family care, discretionary events) is suspended until investigation concludes.

  ◦ Bereavement and jury duty exceptions still apply, but employee must provide live contact details during leave for investigation-related communication.

  ◦ Any leave exceeding 3 days must be approved directly by the Chief HR Officer.
```

Fig 20: Response generated when k=2 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

```

Relevant Chunk 1:
• Jury duty / legal summons: For the full duration specified in the official court document.

• Emergency family care: Up to 3 working days per calendar year.

• Natural disasters: Maximum 7 consecutive calendar days per incident.

• Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements
Special leave is approved only upon submission of:
• Death certificate, funeral notice, or obituary (bereavement).

• Official court summons or jury duty letter (legal obligations).

• Medical certificate from a registered practitioner (emergency care).

• Government-issued disaster report or evacuation notice (natural disasters).

• All documents must be submitted within 5 working days of returning to duty.

4. Impact of Operational Demands
• During peak flight operations (e.g., December 15-January 5 and April 1-10),

```

Fig 21: Response generated when k=2 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

```

Relevant Chunk 1:
• Jury duty / legal summons: For the full duration specified in the official court document.

• Emergency family care: Up to 3 working days per calendar year.

• Natural disasters: Maximum 7 consecutive calendar days per incident.

• Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements
Special leave is approved only upon submission of:
• Death certificate, funeral notice, or obituary (bereavement).

• Official court summons or jury duty letter (legal obligations).

• Medical certificate from a registered practitioner (emergency care).

• Government-issued disaster report or evacuation notice (natural disasters).

• All documents must be submitted within 5 working days of returning to duty.

4. Impact of Operational Demands
• During peak flight operations (e.g., December 15-January 5 and April 1-10),

```

Fig 22: Response generated when k=2 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- The embedding model 'nomic-ai/nomic-embed-text-V1.5' is particularly noteworthy as it successfully retrieved relevant chunks in response to the user's query, whereas the other two embedding models yielded very similar results.

Query 3: What should I do if I notice suspected harassment with my female colleague?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report.
```

Fig 23: Response generated when k=2 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

Relevant Chunk 1:

- Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).
2. Secure Email Portal: hr.ethics@flykiteair.com.
3. Anonymous drop-boxes in crew lounges and staff cafeterias.

- Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
 - Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.
 - Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.
5. Investigation & Resolution Timelines
 - HR must initiate a preliminary review within 3 working days of receiving a report.

Fig 24: Response generated when k=2 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).
2. Secure Email Portal: hr.ethics@flykiteair.com.
3. Anonymous drop-boxes in crew lounges and staff cafeterias.

- Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
 - Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.
 - Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.
5. Investigation & Resolution Timelines
 - HR must initiate a preliminary review within 3 working days of receiving a report.

Fig 25: Response generated when k=2 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

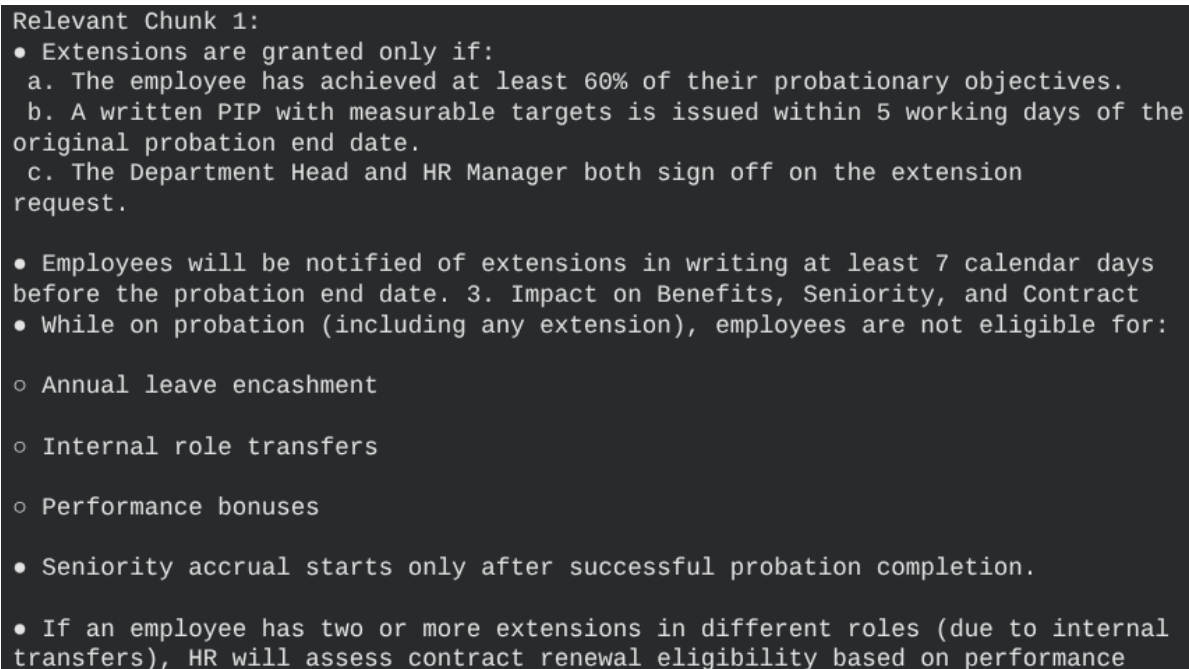
- The ‘nomic-ai/nomic-embed-text-V1.5’ model retrieved information about "Workplace Safety and Health," which was irrelevant to the user's query. In contrast, the other two embedding models retrieved the same, relevant chunks.

Fetching relevant chunks when k=4 for all three embedding models

Query 1: What are the effects on the benefits I receive if my probation is extended?

For ‘thenlper/gte-large’ embedding model

Response:



```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
  c. The Department Head and HR Manager both sign off on the extension request.

• Employees will be notified of extensions in writing at least 7 calendar days before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:

  o Annual leave encashment
  o Internal role transfers
  o Performance bonuses

• Seniority accrual starts only after successful probation completion.

• If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance
```

Fig 26: Response generated when k=4 for ‘thenlper/gte-large’ embedding model

For ‘BAAI/bge-large-en-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Extensions are granted only if:
 - a. The employee has achieved at least 60% of their probationary objectives.
 - b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
 - c. The Department Head and HR Manager both sign off on the extension request.
- Employees will be notified of extensions in writing at least 7 calendar days before the probation end date. 3. Impact on Benefits, Seniority, and Contract
- While on probation (including any extension), employees are not eligible for:
 - Annual leave encashment
 - Internal role transfers
 - Performance bonuses
- Seniority accrual starts only after successful probation completion.
- If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance

Fig 27: Response generated when k=4 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Extensions are granted only if:
 - a. The employee has achieved at least 60% of their probationary objectives.
 - b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
 - c. The Department Head and HR Manager both sign off on the extension request.
- Employees will be notified of extensions in writing at least 7 calendar days before the probation end date. 3. Impact on Benefits, Seniority, and Contract
- While on probation (including any extension), employees are not eligible for:
 - Annual leave encashment
 - Internal role transfers
 - Performance bonuses
- Seniority accrual starts only after successful probation completion.
- If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance

Fig 28: Response generated when k=4 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

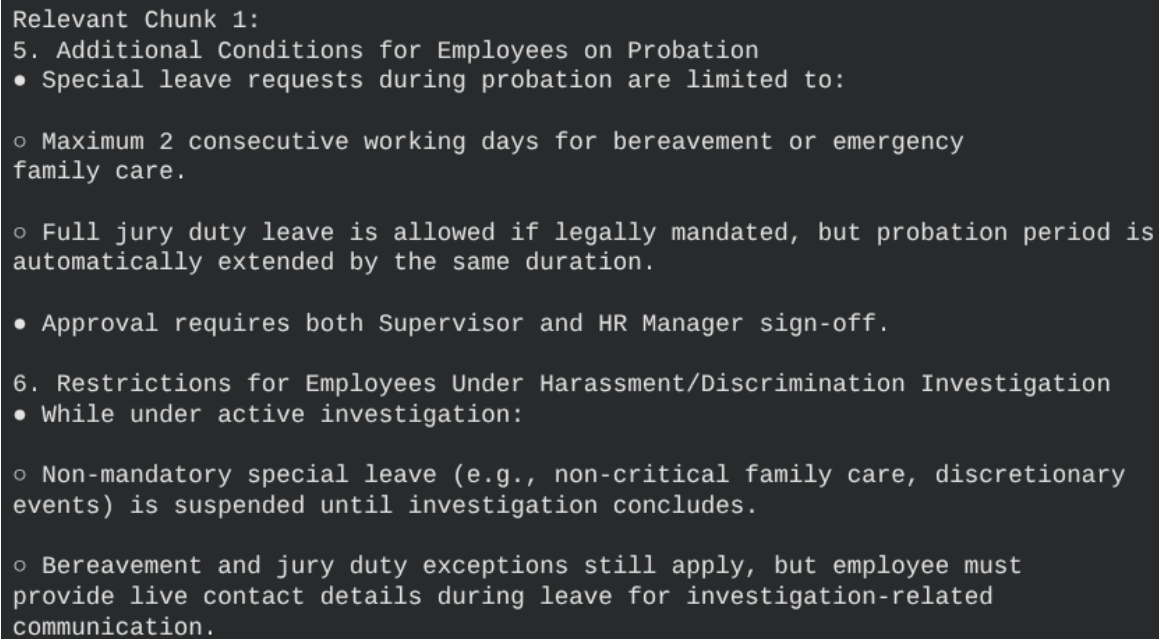
Inference:

- In this case, 'thenlper/gte-large' has not addressed "Payroll and Termination" procedures, whereas the other two models stand out by retrieving appropriate chunks in response to the user query.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

For 'thenlper/gte-large' embedding model

Response:



```
Relevant Chunk 1:
5. Additional Conditions for Employees on Probation
• Special leave requests during probation are limited to:

  ◦ Maximum 2 consecutive working days for bereavement or emergency family care.

  ◦ Full jury duty leave is allowed if legally mandated, but probation period is automatically extended by the same duration.

• Approval requires both Supervisor and HR Manager sign-off.

6. Restrictions for Employees Under Harassment/Discrimination Investigation
• While under active investigation:

  ◦ Non-mandatory special leave (e.g., non-critical family care, discretionary events) is suspended until investigation concludes.

  ◦ Bereavement and jury duty exceptions still apply, but employee must provide live contact details during leave for investigation-related communication.
```

Fig 29: Response generated when k=4 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

Relevant Chunk 1:

- Jury duty / legal summons: For the full duration specified in the official court document.
- Emergency family care: Up to 3 working days per calendar year.
- Natural disasters: Maximum 7 consecutive calendar days per incident.
- Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements

Special leave is approved only upon submission of:

- Death certificate, funeral notice, or obituary (bereavement).
- Official court summons or jury duty letter (legal obligations).
- Medical certificate from a registered practitioner (emergency care).
- Government-issued disaster report or evacuation notice (natural disasters).
- All documents must be submitted within 5 working days of returning to duty.

Fig 30: Response generated when k=4 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Jury duty / legal summons: For the full duration specified in the official court document.
- Emergency family care: Up to 3 working days per calendar year.
- Natural disasters: Maximum 7 consecutive calendar days per incident.
- Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements

Special leave is approved only upon submission of:

- Death certificate, funeral notice, or obituary (bereavement).
- Official court summons or jury duty letter (legal obligations).
- Medical certificate from a registered practitioner (emergency care).
- Government-issued disaster report or evacuation notice (natural disasters).
- All documents must be submitted within 5 working days of returning to duty.

Fig 31: Response generated when k=4 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- Each of the three embedding models retrieved identical chunks that were relevant to the user query, albeit in a different order.

Query 3: What should I do if I notice suspected harassment with my female colleague?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report
```

Fig 32: Response generated when k=4 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

```

Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report.

```

Fig 33: Response generated when k=4 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

```

Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report

```

Fig 34: Response generated when k=4 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- The 'nomic-ai/nomic-embed-text-V1.5' embedding model retrieved data unrelated to the user query, such as 'Data Protection and Privacy' and 'Attendance and Absence Management'. In contrast the other two embedding models retrieved relevant chunks related to the user query.

Fetching relevant chunks when k=8 for all three embedding models

Query 1: What are the effects on the benefits I receive if my probation is extended?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
  c. The Department Head and HR Manager both sign off on the extension request.

• Employees will be notified of extensions in writing at least 7 calendar days before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:
  o Annual leave encashment
  o Internal role transfers
  o Performance bonuses

• Seniority accrual starts only after successful probation completion.

• If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance history. No automatic carry-over is granted.
```

Fig 35: Response generated when k=8 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

Relevant Chunk 1:

- Extensions are granted only if:
 - a. The employee has achieved at least 60% of their probationary objectives.
 - b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
 - c. The Department Head and HR Manager both sign off on the extension request.
- Employees will be notified of extensions in writing at least 7 calendar days before the probation end date. 3. Impact on Benefits, Seniority, and Contract
- While on probation (including any extension), employees are not eligible for:
 - Annual leave encashment
 - Internal role transfers
 - Performance bonuses
- Seniority accrual starts only after successful probation completion.
- If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance

Fig 36: Response generated when k=8 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Extensions are granted only if:
 - a. The employee has achieved at least 60% of their probationary objectives.
 - b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
 - c. The Department Head and HR Manager both sign off on the extension request.
- Employees will be notified of extensions in writing at least 7 calendar days before the probation end date. 3. Impact on Benefits, Seniority, and Contract
- While on probation (including any extension), employees are not eligible for:
 - Annual leave encashment
 - Internal role transfers
 - Performance bonuses
- Seniority accrual starts only after successful probation completion.
- If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance

Fig 37: Response generated when k=8 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

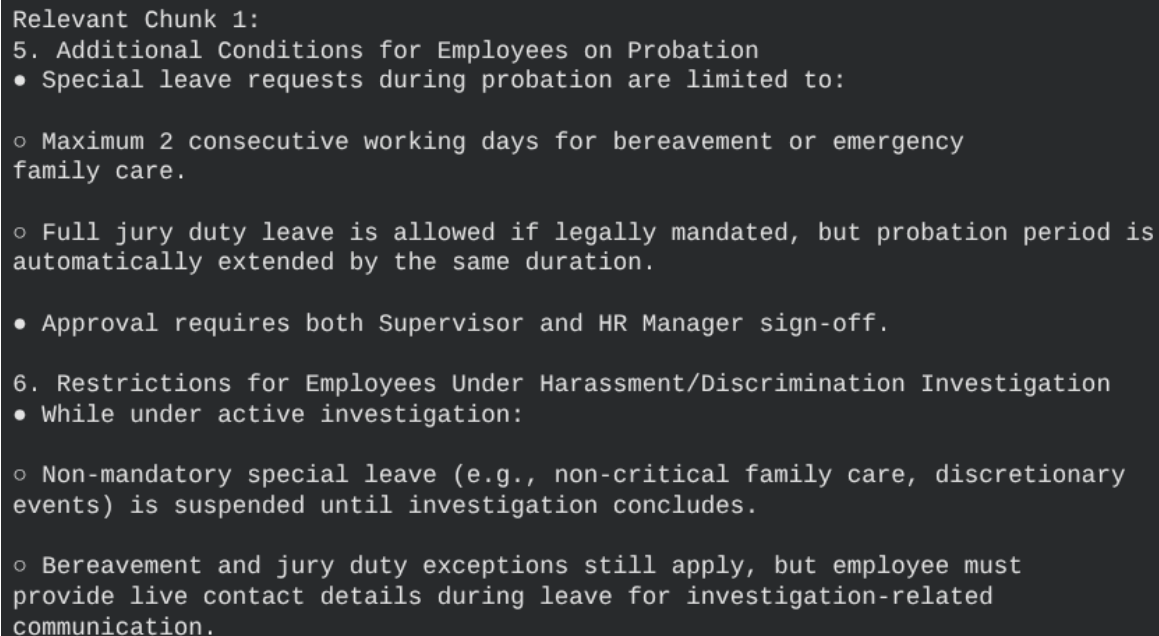
Inference:

- While all of the embedding models were returning the same chunks, some of them were unrelated to the user's inquiry.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

For 'thenlper/gte-large' embedding model

Response:



Relevant Chunk 1:

5. Additional Conditions for Employees on Probation

- Special leave requests during probation are limited to:
 - Maximum 2 consecutive working days for bereavement or emergency family care.
 - Full jury duty leave is allowed if legally mandated, but probation period is automatically extended by the same duration.
- Approval requires both Supervisor and HR Manager sign-off.

6. Restrictions for Employees Under Harassment/Discrimination Investigation

- While under active investigation:
 - Non-mandatory special leave (e.g., non-critical family care, discretionary events) is suspended until investigation concludes.
 - Bereavement and jury duty exceptions still apply, but employee must provide live contact details during leave for investigation-related communication.

Fig 38: Response generated when k=8 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

Relevant Chunk 1:

- Jury duty / legal summons: For the full duration specified in the official court document.
- Emergency family care: Up to 3 working days per calendar year.
- Natural disasters: Maximum 7 consecutive calendar days per incident.
- Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements

Special leave is approved only upon submission of:

- Death certificate, funeral notice, or obituary (bereavement).
- Official court summons or jury duty letter (legal obligations).
- Medical certificate from a registered practitioner (emergency care).
- Government-issued disaster report or evacuation notice (natural disasters).
- All documents must be submitted within 5 working days of returning to duty.

Fig 39: Response generated when k=8 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Jury duty / legal summons: For the full duration specified in the official court document.
- Emergency family care: Up to 3 working days per calendar year.
- Natural disasters: Maximum 7 consecutive calendar days per incident.
- Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements

Special leave is approved only upon submission of:

- Death certificate, funeral notice, or obituary (bereavement).
- Official court summons or jury duty letter (legal obligations).
- Medical certificate from a registered practitioner (emergency care).
- Government-issued disaster report or evacuation notice (natural disasters).
- All documents must be submitted within 5 working days of returning to duty.

Fig 40: Response generated when k=8 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- Regardless of the embedding model employed, a combination of relevant and irrelevant chunks were retrieved based on the user query.

Query 3: What should I do if I notice suspected harassment with my female colleague?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report
```

Fig 41: Response generated when k=8 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

Relevant Chunk 1:

- Accepted reporting channels:

- Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).
- Secure Email Portal: hr.ethics@flykiteair.com.
- Anonymous drop-boxes in crew lounges and staff cafeterias.

- Reports must include date, location, persons involved, and a brief incident description.

- Protection Against Retaliation
 - Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.
 - Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.
- Investigation & Resolution Timelines
 - HR must initiate a preliminary review within 3 working days of receiving a report.

Fig 42: Response generated when k=8 for 'BAAI/bge-large-en-v1.5' embedding model

For 'nomic-ai/nomic-embed-text-v1.5' embedding model

Response:

Relevant Chunk 1:

- Accepted reporting channels:

- Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).
- Secure Email Portal: hr.ethics@flykiteair.com.
- Anonymous drop-boxes in crew lounges and staff cafeterias.

- Reports must include date, location, persons involved, and a brief incident description.

- Protection Against Retaliation
 - Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.
 - Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.
- Investigation & Resolution Timelines
 - HR must initiate a preliminary review within 3 working days of receiving a report.

Fig 43: Response generated when k=8 for 'nomic-ai/nomic-embed-text-v1.5' embedding model

Inference:

- Regardless of the embedding model, every chunk that was retrieved was identical; nevertheless, some irrelevant chunks were also retrieved in accordance with the user query.

Final Inferences:

- Relevant document chunks are retrieved using the "Similarity" search strategy.
- By collecting relevant chunks in response to user queries, the 'BAAI/bge-large-en-v1.5' embedding model performs better.
- Very few contextual pieces were obtained when k=2 was lower. Some unnecessary chunks were obtained when the value of k=8 was higher. In order to ensure that appropriate chunks were obtained in accordance with user queries, any ideal value of k=4 is preferable.

Stage 2 and 3: Question Answering using RAG and Evaluation Findings

To generate a response the LLM model is defined with following parameters i.e., maximum tokens (upper limit on the length of the generated response) to 128, temperature (Controls the randomness or "creativity" of the model's token selection) to 0, Repetition Penalty (Penalizes repeated tokens to improve text diversity) to 1.2, top-p (Accumulates the probabilities of the most likely tokens and discards any tokens outside that top cumulative percentile) to 0.95, top-k (Limits the model's choices to a hard cutoff of the top k most probable next words, completely ignoring the rest of the vocabulary regardless of their probability scores) to 50.

The above parameters were fine-tuned under different scenarios, to get the best parameter combination.

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response:

1. Benefits, such as health insurance and retirement plan enrollment, do not accrue during the probation period.
2. After successful completion of probation, seniority accrual starts for benefit eligibility purposes.
3. If your probation is extended, benefits will continue until the last working day of your employment unless mandated by law otherwise.

Inference:

- The generated context-based response responds to the user's inquiry and functions more like an internal HR bot.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response:

1. Inform your direct supervisor verbally within 6 hours of the incident.
2. Submit written or email notification within 24 hours.

3. Special leave is granted for bereavement up to a maximum of 5 consecutive working days per incident.

Inference:

- It provided quantitative parameters like verbal communication to the supervisor within six hours, email notification within twenty-four hours, and the number of days the vacation can be given, the response is more company-specific.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response:

1. Report the incident using one of the following channels:
 - Confidential HR Helpline: +91-8000-456-789 (available Mon–Sat, 9 AM–8 PM IST)
 - Secure Email Portal: hr.ethics@flykiteair.com
 - Anonymous drop-boxes in crew lounges and staff cafeterias
2. Include the following details when reporting: date, location, persons involved, brief incident description
3. Protection against retaliation applies to you as well; report

Inference:

- The response is extremely domain-specific, effectively incorporating precise company policy metrics that were obtained from the source context. It also provides an internal HR hotline number and an email address so that an employee can send a notification. The answer is also incomplete.

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated Result:

```
Groundedness Evaluation
5: The metric is followed completely.
[EXPLANATION]: The answer strictly adheres to the information provided in the context. It states that employees do not accrue benefits

Relevance Evaluation
5: The metric is followed completely.
[EXPLANATION]: The AI-generated answer directly addresses the main aspects of the question regarding the effects on benefits during and
```

Fig 44: Evaluated Response generated by RAG model for user query 1

Inference:

- Since the response is entirely derived from the context, the groundedness score was a perfect 5.
- By fully addressing the key elements of the user's question, the generated response fully satisfies the metric requirement for relevance, earning a perfect score of 5.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated Result:

```
Groundedness Evaluation
Based on the provided context, here are the steps needed to verify the truthfulness of the answer:

1. The employee should inform their direct supervisor verbally within 6 hours of the incident as stated in the context under "Notification Process".
2. Within 24 hours, they must submit written or email notification about the leave request as mentioned in the same section.
3. According to the "Special Leave Policy – Flykite Airlines," special leave is granted for bereavement up to a maximum of 5 consecutive days.

The generated answer matches the context, as it includes both steps required by the company policy for reporting an emergency leave request.
Therefore, [3] - The metric is followed to a good extent.

Relevance Evaluation
The AI-generated answer addresses the following core parts of the question:
1. Informing the direct supervisor about the leave due to family demise.
2. Submitting written or email notification within the specified time frame.
3. Determining the duration limit for bereavement special leave.

The AI-generated answer directly references the context provided, specifically points 1 (Notification Process), 4 (Special Leave Policy), and 5 (Bereavement Leave).
Therefore, based on the given evaluation criteria, I would rate this AI-generated answer as: [5] - The metric is followed completely.
```

Fig 45: Evaluated Response generated by RAG model for user query 2

Inference:

- Since the metric is mostly adhered to, the evaluation of groundedness yielded a score of 3.
- Because the response adheres to the user's inquiry by covering nearly every area, the evaluation of relevancy yielded a flawless 5.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated Result:

```
Groundedness Evaluation
[5]: The answer is rated a 5 because every single fact in the answer is explicitly backed by the context provided.

[EXPLANATION]: The AI-generated answer strictly adheres to the information presented in the context. It accurately reflects the steps and channels mentioned for reporting suspected harassment.

Relevance Evaluation
To evaluate the answer's relevance to the question, follow these steps:
1. Identify the main aspects of the question: a) Noticing suspected harassment with a female colleague and b) Reporting it.
2. Check if each aspect is addressed in the answer:
  a) The answer instructs reporting using accepted channels (confidential HR Helpline, secure email portal, or a formal written report).
  b) However, the first part of the question asks what should be done if one notices suspected harassment with a female colleague.
3. Evaluation:
Based on the given evaluation criteria, I would rate this answer as a 2 because while the answer does provide
```

Fig 46: Evaluated Response generated by RAG model for user query 3

Inference:

- Since the whole response is taken from the context, the evaluation score for groundedness is a perfect 5.
- The response produced does not specifically address what should be done if one witnesses suspected harassment before reporting it, earning the evaluation score for relevance of 2.

Conclusion

- When contrasted with the answers produced by prompt engineering, RAG, and llm. Since the context is exposed to the llm in order to respond to an employee inquiry, the responses produced by the RAG model were encouraging.
- The examination yielded scores of 3 and 5 when every response was checked for groundedness. Because the model overlooked important policy boundaries, prerequisites, or caveats in one response, two responses were inferred from the context.
- The evaluation came out as 2 and 5 after every response was checked for relevancy. Due to the fact that two responses fully addressed the user's question, one response did not.
- The groundedness score may be improved by fine-tuning the hyper-parameters, which may then be tested.

Fine-tuning (Case-1): max_tokens=256, temperature=1, top_p=0.5, top_k=90, repeat_penalty=0
Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response:

if the. prob fileH file on2 benefits by forH@ for/ for@ benefits/v only file2? benefits I [isv only/ effects

@ extended my extended extended effects the2 receive are the isv only are@ for prob?H prob only [benefits by@v if if.H if [receive I. effects on onv file extended/v effects [? I receive@ I is is benefits for file file on@ for on if2 extended the receive my I are. file extended benefits/ on2 for prob effects@ is [H for effects I prob extended by benefits prob@ the effects on receive my/ on receivev on is2 for is file by. I// I prob [... if is [for for2 by probv is file byH by benefits [2 benefits myv2 my prob by by effectsv effects/HH receiveH by if benefits/ by if for on for [my for file benefits/ by [v/ benefits effects benefits forv effects myv file effects2 if isH forH receive for by [benefits/ for receive is on fileH by file is is2vv is is [receive by by [is is effects on by effects my by by on

Inference:

- With excessive token repetition and randomization, the answer is unrelated to the user's question.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response:

inform inform office inform3v last/?3 only family only. need last has attend night office has should inv3 inform [v need will.v will r familyv night office only?? How should leave office3 be only? only3 in office r attend will should. attend be How office attend How last3 r be has3 should will should office [night. office hasv in in in be last should night inform night only inform office be How family3v office inform leave bev be be attend has night be3 has. family last last office be last in family be office leave attend onlyvv will r only How in in attend in in leave office [in only be

attend. leave family be last only r be office How night office night will inv family How [be. only
 How night only3. leave r [v only only office night family night [last3 be only r [.. only leave only
 willv leave family r in r be leave night officev night family last bev How. r leave How nightv r
 How3 leavev office. in in will3 in last3 in in leave leave3 invv night family r nightv office r [How
 r in3 r.3 night last last will

Inference:

- When compared to the user inquiry, which has a significant degree of randomness and token repetition, the generated response is meaningless and lacks contextual relevance.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response:

file is by [ine with is/ with if/ I pen pen.ch my meant my harass use file/ do@ female file do3 is
 my notice only harass pen/ female3ine with is I female with is suspected is colle with only colle
 by colleinech only/ with suspected [do use suspected file female only. my pen bych@ pench I [
 my3@ pen suspected harass3 harass by pench file. by suspected@ is do colle@ I harass pen
 collech@33 mych my3 do use suspected. I. suspected@ine@ is do use3 harass is useineine I@ch
 by by my by is [do@ is pen [pen. do harass is by harass suspected is colle colle by harass harass
 colle do pen use3 doch@ collech. by3inech by3 suspected I.. I pen suspected harass I harass doch
 do@ by suspected colle pen3ine by my myine suspected by I by colle by [ine colle3 usech [ine my
 is my I I usech suspected is I by doch use. is harass suspected colle@ [useine pen my harass my
 harass pen is do my I [by by bych use [use [ine use pen.

Inference:

- Extreme token looping and a complete lack of grammatical or contextual reasoning characterize this utterly irrelevant and pointless response.

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated Result:

```
Groundedness Evaluation
Based on the provided context, it's not possible to accurately determine an answer to the question due to the garbled and

[0]: This rating signifies that no evaluation can be made based on the provided information.
[The given answer is not aligned with the question or context, making it impossible to evaluate its groundedness.]

Relevance Evaluation
To evaluate the answer, I will first identify the main aspects of the question and then check how well they are addressed

1. What happens to benefits when probation is extended?
2. Are there any specific rules or conditions for receiving benefits during an extension period?

Now, let's examine the answer and see if it covers these aspects:

The AI-generated response does mention "benefits" several times but lacks a clear connection to the question about probati
Based on this analysis, I would rate the relevance of the AI-generated response as 1 (The metric is not followed at all).
```

Fig 47: Evaluated Response generated by RAG model for user query 1

Inference:

- The score for groundedness was zero since the generated response is meaningless, has many faults, and is not derived from the context.
- Because the generated response does not address the key elements of the user inquiry, it does not meet the metric criterion for relevance, resulting in a score of 1.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated Result:

```
Groundedness Evaluation
Based on the provided context, it is not possible to determine an accurate and grounded answer for the user's question about i

1. Identify specific claims made in the generated answer that can be aligned with information from the context.
2. Determine if each claim is strictly derived from the provided context or introduces outside knowledge, assumptions, or extrap
3. Evaluate the extent to which the metric of groundedness is followed based on the alignment check and explicit steps above.

However, due to the lack of clear claims in the generated answer that can be directly linked with information from the context,

[0]: No rating provided as there is no valid answer to evaluate based on the context.

Relevance Evaluation
1: The AI-generated answer does not follow the metric at all as it is difficult to understand and does not directly address the
```

Fig 48: Evaluated Response generated by RAG model for user query 2

Inference:

- Since the response is not based on the context, the groundedness score is 0.
- Since the response does not follow the user's inquiry by offering some random tokens, the score obtained for evaluating relevancy is 1.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated Result:

```
Groundedness Evaluation
Based on the provided context, there isn't a clear and explicit answer to the question regarding what actions should be

To verify the truthfulness of any potential AI-generated answers, we would need to:
1. Identify specific actions recommended in the answer based on the information provided in the context.
2. Check if each action is strictly derived from the context without introducing assumptions or outside knowledge.
3. Ensure that no extrapolations are made beyond what's stated in the context.

However, since there isn't a clear and explicit answer in the context regarding specific actions to take when noticing su

[1]: The metric is not followed at all.
[EXPLANATION]: The provided context does not contain enough information for generating a strictly and exclusively derived

Relevance Evaluation
To evaluate the relevance of the AI-generated answer to the question, we need to identify if it provides steps or advice

1. Identify keywords from both the question and context that can help guide our evaluation:
- Suspected harassment
- Female colleague
- Reporting channels (HR Helpline, Secure Email Portal, Anonymous drop-boxes)
- Protection against retaliation

2. Break down the AI-generated answer into meaningful parts and compare them with the keywords identified:
a. "file is by [ine with is/ with if/ I pen pen.ch my meant my harass use file" - This part does not seem to contain a
b. "[ do@ female file do3 is my notice only harass pen/ female3ine with is I female with is suspected is colle with or
c. "[ do use suspected harass is useineine I@ch by by is [ do@ is pen [ pen. do harass is by harass suspected is colle

Based on this analysis, it appears that the AI-generated answer does not directly address the main aspects of the question

[1]: This answer does not follow the relevance metric at all, as it fails to address the main aspects of the question by
```

Fig 49: Evaluated Response generated by RAG model for user query 3

Inference:

- Since the generated response is meaningless and nearly fails to take information from context, the evaluation score for groundedness is 1.
- The evaluation score for relevance is 1. The generated response does not follow the user's question; it does not address key points and does not give a clear explanation.

Conclusion

- The aforementioned RAG model repeatedly failed to offer pertinent answers to the employee's user concerns.
- Because the repeat penalty parameter is set to 0, the response produced by the RAG model contained a token repetition.
- Additionally, the model produced answers with a significant degree of randomness; this could be due to the temperature parameter being set to 1.
- Because the top_p parameter value is set to 0.5, the model is unable to produce a variety of replies.
- The evaluation ranged from 0 to 1 when all of the responses were checked for groundedness because the inconsistent parameter values prevented the respondent from producing answers based on context.
- The evaluation came out to be 1 after every response was checked for relevancy, because the generated responses did not fully follow the user's inquiry.
- The groundedness and relevance score may be enhanced by further hyper-parameter tuning, which can be tested later.

Fine-tuning (Case-2): max_tokens=512, temperature=0.8, top_p=0.8, top_k=70, repeat_penalty=0.5

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response:

effects on benefits for probation extended employees

1. benefits are only eligible for employees on probation if probation is successfully completed
2. if probation is extended, benefits only continue until the extended probation ends
3. benefits, if eligible, are only eligible for employees on probation if the probation is successfully completed, the extended probation is successfully completed,

Inference:

- Only a portion of the answer given by the model is relevant to the user inquiry, and several tokens were redundant and insignificant.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response:


```

Llama.generate: prefix-match hit
I. How to inform the office

* Verbal notification to the direct supervisor within 6 hours.
* Written/email notification within 24 hours.

II. leave granted

* granted for the death of the employee's immediate family.
* for the death of the employee's immediate family.

II. leave granted

* granted for the death of the employee's immediate family.
* for the death of the employee's immediate family.

II. leave granted

* granted for the death of the employee's immediate family.
* for the death of the employee's immediate family.

* Up to for the death of the employee's immediate family.

```

Fig 50: Response generated by RAG model for user query 2

Inference:

- The model's response to the user query is pertinent, but a few phrases were repeated.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response:

```

Llama.generate: prefix-match hit
I. Report suspected harassment by using one of the following channels:

1. Confidential HR Helpline: +91-8000-456-789
2. Secure Email: HR.ethics@flykiteair.com
3. Anonymous Report: Anonymous Report-box

II. Report must include:

- Date
- Location
- Persons involved
- Brief incident description

II. Protection:

- Reporting employee: Protection against retaliation
- Report retaliation: Report retaliation against employee within 7-calendar-days

II. Reporting:

- Reporting: Report within 15-calendar-days
- Reporting: Reporting employee: Reporting employee retaliation within 7-calendar-days

```

Fig 51: Response generated by RAG model for user query 3

Inference:

- The answer is significant to the user's query; yet, a few phrases are unrelated and repetitive.

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated Result:

```
Groundedness Evaluation
I. Evaluation of the generated answer for groundedness with respect to the context

1. Verification steps:
a. Identify if all claims in the answer are explicitly stated in the context.
b. Check if any assumptions or extrapolations have been made in deriving the answer.
c. Ensure that no outside knowledge has been introduced into the answer.

2. Alignment check:
a. The first claim "benefits are only eligible for employees on probation if probation is extended" is explicitly stated in the context.
b. The second claim "if probation is extended, benefits only continue until the extension period" is explicitly stated in the context.
c. The third claim "benefits, if eligible, are only eligible for employees on probation" is explicitly stated in the context.

3. Groundedness evaluation:
The generated answer follows the metric to a good extent (rating: 3). While some of the claims are explicitly stated in the context, others require inference based on numerous points in the context.

Relevance Evaluation
I. Evaluation of AI-generated answer for question about the effects on benefits for employees on probation if probation is extended.

1. Understanding the Question: The user is asking about the specific impact on benefits if probation is extended.
2. Context Analysis: The context provided outlines various aspects related to probation and benefits.
3. AI-generated Answer Explanation: The answer states that benefits are only eligible for employees on probation if probation is extended, and that benefits only continue until the extension period.
4. Relevance Evaluation: While the answer does touch upon the main aspects of the question, it does not provide a comprehensive overview of the effects on benefits.
5. Final Evaluation: [2] The AI-generated answer follows the metric to a limited extent, addressing only a few parts of the user's query.
```

Fig 52: Evaluated Response generated by RAG model for user query 1

Inference:

- The evaluation score for groundedness is 3, because certain parts of the response are generated by the context, while others require inference based on numerous points in the context.
- The relevance evaluation score is 2, and the generated response follows the metric criteria to a limited extent, addressing only a few parts of the user's query.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated Result:

```
Groundedness Evaluation
[5]: The metric is followed completely. Every single fact in the answer is explicitly backed by the context.

[EXPLANATION]: The AI-generated answer strictly adheres to the information presented in the context.

Relevance Evaluation
I. Evaluation Steps:
1. Identify if the answer covers both parts of the question: informing the office and leave granting.
2. Check if each part is addressed correctly based on the policy provided in context.
3. Ensure that all important aspects of the question are included in the answer.

II. Explanation:
The AI-generated answer covers both parts of the question, informing the office and leave granting. For the first part of the question regarding informing the office, the answer states that the user should inform the office immediately. For the second part of the question regarding leave granting, the answer states that specific leave policies should be followed.

III. Evaluation:
[3] - The metric is followed to a good extent as the AI-generated answer covers both parts of the question.
```

Fig 53: Evaluated Response generated by RAG model for user query 2

Inference:

- The score for measuring groundedness is 5, indicating that the created response is contextually appropriate.
- The resulting score for measuring relevance is 3, indicating that the metric is well-followed while addressing only a few aspects of the user question.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated Result:

```
Groundedness Evaluation
[5]: The metric is followed completely. Every single fact in the answer is explicitly based on the context.
[EXPLANATION]: The AI-generated answer strictly adheres to the information provided in the context.

Relevance Evaluation
1. To evaluate the answer, we need to check if it addresses all important aspects of the user query.

1. Reporting Channels: The answer correctly lists three ways to report suspected harassment.
2. Required Information for Reports: The answer includes the necessary information that should be provided when reporting.
3. Protection Against Retaliation: The answer mentions protection against retaliation for the reporting employee.
4. Reporting Timeline: Although the answer does not explicitly state a specific timeline, it implies that reporting should be done as soon as possible.
5. Investigation and Resolution Process: The answer briefly outlines some aspects of the process, such as the investigation and the possibility of disciplinary actions.
6. Disciplinary Actions for Violations: Although the answer mentions possible outcomes for the perpetrator, it does not provide specific details about the disciplinary actions.
7. Overall Relevance: The answer addresses most of the important aspects mentioned in both the query and the context.

[3]: The metric is followed to a good extent as the AI-generated answer covers essential aspects of the query.
```

Fig 54: Evaluated Response generated by RAG model for user query 3

Inference:

- The evaluation score for groundedness is 5, indicating that the metric is completely followed, with accurate contact information and process.
- The relevance evaluation score is 3, because the result closely matched the user query, although a few tokens were irrelevant.

Conclusion

- The above RAG model's answers have improved when compared to those of the prior model. However, a few words and tokens were repeated, and the format was not very regular.
- The model's answer to user queries contains some token repetition because the repeat penalty parameter is set to 0.5.
- When compared to the prior model, the randomness of the generated response has been marginally enhanced by setting the temperature parameter to 0.8.
- The model has greater response variety than the prior scenario since the top_p parameter is set to 0.8.
- When all of the responses were evaluated for groundedness, the results were 5 and 3, indicating that two of the responses were mostly drawn from the context, while one response required multiple points from the context.

- When all of the responses were considered for relevance, the evaluation yielded 2 and 3. Since the generated responses only partially adhered to the user's inquiry, they did not address the entire user query.
- Further fine-tuning of hyper-parameters may increase the score for groundedness and relevance, which can be tested later.

Fine-tuning (Case-3): max_tokens=512, temperature=0.5, top_p=0.8, top_k=80, repeat_penalty=0.8

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response:

```
Llama.generate: prefix-match hit
I. Probation Extension and Benefits

A. Probation Extension Conditions

1. Achieved at least 60% of probationary objectives
2. A written PIP is issued
3. Both Department Head and HR Manager sign off

B. Benefits during Probation Extension

1. Annual leave encashment is not allowed
2. Internal role transfers are not allowed
3. Performance bonuses are not allowed
4. Seniority accrual starts only after successful probation completion

C. Benefits after Probation Extension

1. Benefits end on last working day unless law mandates otherwise
2. Employees are not eligible for benefits during probation extension
```

Fig 55: Response generated by RAG model for user query 1

Inference:

- The answer provided by the model addresses the user query, although there are a few sentences that are unrelated.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response:

```

Llama.generate: prefix-match hit
* Inform your direct supervisor verbally within 6 hours of the incident.
* Submit a death certificate, funeral notice, or obituary within 5 working days of returning to duty.
* Special leave of up to 5 consecutive working days will be granted for bereavement.
* Full-time employees are entitled to up to 5 days of bereavement leave per incident.
* The leave will be granted, provided the necessary documentation is provided.
* The leave will be deducted from the employee's annual leave entitlement.
* The employee's annual leave balance will be adjusted accordingly.
* The employee's annual leave balance will be communicated by the HR department.
* The employee's leave balance will be available on the HR portal.
* The employee will be able to check their leave balance by logging into the HR portal.
* The employee will be able to check their leave balance by contacting the HR department.
* The employee will be able to contact the HR department by emailing HR@flykiteairlines.com.
* The employee will be able to contact the HR department by calling the HR helpline at 1-800-123-4567.
* The employee will be able to contact the HR department by contacting their HR business partner.
* The employee will be able to contact their HR business partner by emailing their HR business partner's
* The employee will be able to contact their HR business partner by calling their HR business partner's
* The employee will be able to contact their HR business partner by contacting their HR business partner's

```

Fig 56: Response generated by RAG model for user query 2

Inference:

- Certain aspects in the response addressed the customer query, while others were repeated and irrelevant.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response:

```

1. Report the suspected harassment using the following channels:
  - Confidential HR Helpline: +91-8000-456-789
  - Secure Email Portal: hr.ethics@flykiteair.com
  - Anonymous drop-boxes in crew lounges and staff cafeterias

2. Reports must include:
  - Date
  - Location
  - Persons involved
  - Brief incident description

3. Protection against retaliation:
  - Report retaliation against the reporting employee within 7 calendar days.
  - HR will take immediate disciplinary action against retaliation.

4. Investigation & Resolution:
  - HR will initiate a preliminary review within 3 days.
  - Formal investigation will begin within 7 days.
  - HR will provide a written acknowledgment within 48 hours.
  - HR will provide a written outcome summary within 5 days.

```

Fig 57: Response generated by RAG model for user query 3

Inference:

- The generated answer addresses the user's query, contains a few unrelated points, and has some formatting issues.

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated Result:

```
Groundedness Evaluation
[5]: The answer strictly adheres to the context provided, as all facts stated in the

[EXPLANATION]: The AI-generated answer accurately reflects the conditions for probati

Relevance Evaluation
I. Evaluation of AI Answer: Effects on Benefits if Probation is Extended

A. Explicit evaluation steps:
1. Identify the main aspects of the question regarding benefits during and after prob
2. Check whether the answer covers all these aspects in detail.
3. Assess how accurately and comprehensively each point is addressed by the AI-genera
4. Evaluate if any additional information provided by the context influences the rele

B. Step-by-step explanation:
1. The question asks about the effects on benefits when probation is extended, focusi
2. The AI-generated answer covers all these aspects in detail under sections B (Benef
3. Each point is addressed accurately, stating that annual leave encashment, internal
4. The context does not provide any additional information that would influence the r

C. Overall evaluation:
The AI-generated answer follows the metric completely (5) as it directly, fully, and
```

Fig 58: Evaluated Response generated by RAG model for user query 1

Inference:

- The score for groundedness was 5, and the metric is totally followed by generating responses from context.
- For relevance, the generated response totally followed the metric requirements, resulting in a score of 5, indicating that key aspects were addressed appropriately.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated Result:

```
3: The metric is followed to a good extent

Explanation:
The AI-generated answer primarily relies on the context provided, which includes information

Relevance Evaluation
5: The metric is followed completely.
[EXPLANATION]: The AI-generated answer directly addresses the main aspects of the question,
```

Fig 59: Evaluated Response generated by RAG model for user query 2

Inference:

- The score obtained for measuring groundedness is 3, because several claims in the generated response go beyond the context.

- The resulting score for evaluating relevance is 5, indicating that the generated response perfectly adheres to the user query.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated Result:

```

Groundedness Evaluation
[3]: The answer follows the metric to a good extent as it primarily relies on the provided context.

[EXPLANATION]: The generated answer is primarily derived from the provided context and only includes information that is present in the context.

To verify the truthfulness of the answer against the provided context, you can follow the steps below:

1. Check if all reporting channels listed in the answer are present in the context (Confidentiality, HR, Legal, etc.).
2. Verify that the required report content matches what's stated in the context (date, location, etc.).
3. Confirm protection against retaliation is consistent with the context (reporting without fear of retaliation).
4. Ensure investigation & resolution timelines align with the context (preliminary review, investigation, resolution).
5. Verify that disciplinary actions for violations are consistent with what's stated in the context (warning, suspension, termination).
6. Check if equal opportunity policy is consistent between the answer and the context (zero-tolerance policy for gender harassment).

Relevance Evaluation
[3]: The AI-generated answer follows the metric to a good extent. It directly addresses the user's query and provides a clear, concise response.

```

Fig 60: Evaluated Response generated by RAG model for user query 3

Inference:

- For groundedness the evaluation score is 3, since the measure is followed to a good extent, but, several lines—like the zero-tolerance policy for gender harassment—were repeated in the response without explicitly acknowledging their original source in the context.
- The relevance evaluation score is 3, because the response does not address details regarding the implications and might be improved.

Conclusion

- When contrasted to the prior scenario, the model's responses are relevant and address the majority of the user query.
- The temperature parameter was changed to 0.5, which reduced the randomness noticed in the model's reactions when compared to earlier models.
- When all of the responses were evaluated for groundedness, the results were 3 and 5. While one response entirely complies to the user inquiry, the other two either go beyond the context or repeat sentences that do not explicitly mention the context.
- When all of the responses were evaluated for relevance, the results were 3 and 5, with two responses adhering to the user inquiry and one failing to address a few areas of the user query.
- Further fine-tuning of hyper-parameters may increase the score for groundedness and relevance, which can be tested later.

Fine-tuning (Case-4): max_tokens=1024, temperature=0.2, top_p=0.95, top_k=80, repeat_penalty=1

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response:

```
* Benefits, such as health insurance and retirement plan, only start accruing after successful probation
* During the probation period (including extensions), employees are not eligible for annual leave encashment
* Seniority accrual also starts only after successful probation completion.
* If an employee has two or more extensions in different roles, HR will assess contract renewal eligibility
* There is no information provided regarding the impact of probation extensions on specific benefits, such as
I don't know.
```

Fig 61: Response generated by RAG model for user query 1

Inference:

- The model's response directly adheres to the user inquiry; yet, the presence of the statements "There is no information provided" and "I don't know" causes a structural contradiction in the user experience and disrupts the intended operational persona.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response:

```
Llama.generate: prefix-match hit
* Inform your direct supervisor verbally within 6 hours of the incident.
* Submit a written or email notification within 24 hours.
* Special leave is granted for bereavement up to 5 consecutive working days per incident.
* Ensure that the documentation (death certificate, funeral notice, or obituary) is submitted within 5 working days
```

Fig 62: Response generated by RAG model for user query 2

Inference:

- The generated response adheres to the user query, addressing most of the aspects.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response:

```
Llama.generate: prefix-match hit
1. Report the incident to the Confidential HR Helpline: +91-8000-456-789 or use the Secure Email Port
2. Provide details such as date, location, persons involved, and a brief description of the incident.
3. Protect your colleague from retaliation by reporting any alleged retaliation within 7 calendar days
4. HR will initiate a preliminary review within 3 working days and begin a formal investigation within
5. Both the complainant and accused will receive written acknowledgment of the complaint within 48 hours
6. Possible outcomes for confirmed violations include written warnings, mandatory sensitivity and work
7. If you are unsure, contact the HR department for clarification.

I don't know: penchala.vineeth3@gmail.com
R02IHZ4C3E
This file is meant for personal use only.
```

Fig 63: Response generated by RAG model for user query 3

Inference:

- The model's response adheres to the user's query. Almost supplying every detail, but failing to function as an HR persona because the response included "I don't know" and a few irrelevant lines.

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated Result:

```
Groundedness Evaluation
[1]: The answer does not strictly follow the context since it makes a claim about the potential

Relevance Evaluation
[3]: The AI-generated answer follows the metric to a good extent by addressing the main aspects

[EXPLANATION]: The AI-generated answer acknowledges that benefits like health insurance and retir
```

Fig 64: Evaluated Response generated by RAG model for user query 1

Inference:

- The groundedness evaluation score is 1, because the response included assumptions and outside knowledge about the impact on health insurance and retirement plans that were not explicitly articulated in the presented context.
- The evaluation score for relevancy is 3, and the generated response may have been enhanced by addressing a few additional aspects in comparison to the user inquiry.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated Result:

```
Groundedness Evaluation
[3]: The answer follows the metric to a good extent as it directly references and a

1. Verbal notification within 6 hours is mentioned in the context for emergencies bu
2. The documentation submission deadline (within 5 working days of returning to duty
3. The answer does not mention any limitations on the number of consecutive bereaver

[EXPLANATION]: The answer follows most of the information from the provided context

Relevance Evaluation
The answer provided by the AI system follows the metric to a good extent (3). Here

1. Explicit steps for evaluation:
  a. Verify if the answer covers all aspects of informing the office and applying f
  b. Check if the documentation requirements are mentioned, including submission ti

2. Addressing core parts of the question:
  The AI-generated answer explicitly states that the employee should inform their o

3. Overall evaluation: [3]
  The answer covers all essential aspects mentioned in the question, including noti

[3]: The answer follows the metric to a good extent as it covers all essential aspe
```

Fig 65: Evaluated Response generated by RAG model for user query 2

Inference:

- The score for measuring groundedness is 3, indicating that the generated response adheres to the context to a significant extent, but there are a few assumptions.
- The resulting score for evaluating relevance is 3, indicating that the measure is being followed to a good extent in addressing a few areas of the user question.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated Result:

```
Groundedness Evaluation
[5]: The answer follows the metric completely as every single fact in the answer is explicitly backed
[EXPLANATION]: The AI-generated answer strictly adheres to the information given in the context, prov

Relevance Evaluation
[5]: The AI-generated answer directly and comprehensively addresses the question by providing specif
[EXPLANATION]: The AI-generated answer directly responds to the user's question by providing a clear
```

Fig 66: Evaluated Response generated by RAG model for user query 3

Inference:

- For groundedness, the evaluation score is perfect 5, as the provided responses is contextually appropriate.
- The relevancy evaluation score is 5, indicating that the response entirely adheres to the user question.

Conclusion

- The model has greatly improved when compared to the previous scenario, addressing the majority of the user query.
- The model's response was devoid of token repetition since the repeat penalty parameter was set at 1.
- The significant decrease in response randomness, as compared to the prior model, is directly due to setting the temperature parameter to a relatively low value of 0.2.
- When all of the responses were assessed for groundedness, the results were 1, 3, and 5. Because one response is entirely taken from the context, the other two responses demonstrate contradictions by providing responses based on assumptions.
- When all of the responses were considered for relevance, the evaluation yielded 3 and 5. Only one response totally corresponded to the user question, while the other two responses contained discrepancies and failed to address certain parts of the query.
- Further fine-tuning of hyper-parameters may increase the score for groundedness and relevance, which can be tested later.

Fine-tuning (Case-5): `max_tokens=1024, temperature=0, top_p=0.95, top_k=50, repeat_penalty=1.2`

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response:

- Benefits, such as health insurance and retirement plan enrollment, do not accrue during the probation period.
- After successful completion of probation, seniority accrual starts for benefit eligibility purposes.
- If your probation is extended, benefits will continue until the last working day of your employment unless mandated by law otherwise.

Inference:

- The answer generated by the model addressed the user query by discussing every aspect.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response:

- Inform your direct supervisor verbally within 6 hours of the incident.
- Submit written or email notification within 24 hours.
- Special leave is granted for bereavement up to a maximum of 5 consecutive working days per incident.

Inference:

- The response created addressed the user query correctly, describing the entire procedure and acting more like an HR persona.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response:

1. Report the incident using one of the following channels:
 - Confidential HR Helpline: +91-8000-456-789 (available Mon–Sat, 9 AM–8 PM IST)
 - Secure Email Portal: hr.ethics@flykiteair.com
 - Anonymous drop-boxes in crew lounges and staff cafeterias
2. Include the following details when reporting: date, location, persons involved, brief incident description
3. Protection against retaliation applies to you as well; report any retaliatory actions within 7 calendar days of occurrence.

Inference:

- The model's response accurately addressed the user query.

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated Result:

```

Groundedness Evaluation
5: The metric is followed completely.
[EXPLANATION]: The answer strictly adheres to the information provided in the context. It states that

Relevance Evaluation
5: The metric is followed completely.
[EXPLANATION]: The AI-generated answer directly addresses the main aspects of the question regarding

```

Fig 67: Evaluated Response generated by RAG model for user query 1

Inference:

- For groundedness, the score was 5, indicating that the measure is completely followed, with the response drawn from context.
- For relevance, the generated response entirely followed the measure, earning a score of 5, addressed every part of the query.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated Result:

```

Groundedness Evaluation
Based on the provided context, here are the steps needed to verify the truthfulness of the answer:

1. The employee should inform their direct supervisor verbally within 6 hours of the incident as stated in the context under
2. Within 24 hours, they must submit written or email notification about the leave request as mentioned in the same section
3. According to the "Special Leave Policy – Flykite Airlines," special leave is granted for bereavement up to a maximum of

The generated answer matches the context, as it includes both steps required by the company policy for reporting an emergency.
Therefore, [3] - The metric is followed to a good extent.

Relevance Evaluation
The AI-generated answer addresses the following core parts of the question:
1. Informing the direct supervisor about the leave due to family demise.
2. Submitting written or email notification within the specified time frame.
3. Determining the duration limit for bereavement special leave.

The AI-generated answer directly references the context provided, specifically points 1 (Notification Process), 4 (Special Leave Policy), and 3 (Bereavement Leave).
Therefore, based on the given evaluation criteria, I would rate this AI-generated answer as: [5] - The metric is followed completely.

```

Fig 68: Evaluated Response generated by RAG model for user query 2

Inference:

- The score for measuring groundedness is 3, indicating that the metric was followed to a good level.
- The resulting relevance score is 5, indicating that the generated response perfectly adheres to the user question by addressing all aspects.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated Result:

Groundedness Evaluation
 [5]: The answer is rated a 5 because every single fact in the answer is explicitly backed by the

[EXPLANATION]: The AI-generated answer strictly adheres to the groundedness metric as it only inc

Relevance Evaluation
 To evaluate the answer's relevance to the question, we need to check if it addresses all importa

1. What should I do if I notice suspected harassment with my female colleague?
2. Accepted reporting channels and required information for reports.
3. Protection against retaliation.

The answer directly addresses these points by suggesting three ways to report the incident, empha

[5]: The metric is followed completely. The answer directly, fully, and comprehensively resolves

Fig 69: Evaluated Response generated by RAG model for user query 3

Inference:

- The evaluation score for groundedness is 5, indicating that the criteria is entirely followed by providing an answer based on context.
- The relevancy evaluation score is 5, as the response adheres to the user inquiry and addresses all aspects.

Conclusion

- When compared to earlier instances, the model's responses are relevant and address all aspects of the user query.
- When all of the responses were evaluated for groundedness, the results were 3 and 5, indicating that the responses were primarily derived from the context, with only one response demonstrating minor discrepancies.
- When all of the responses were evaluated for relevancy, the outcome was a 5. All of the responses were perfectly relevant to the user query.
- This shows that the response is contextually appropriate and related to the user's question. Therefore, both retrieval and augmentation are excellent.
- The combination of perfect groundedness and relevance ensures that the user obtains an answer that is both correct (grounded) and useful (relevant), hence increasing the utility of the RAG system.

Insights and Recommendations

Insights

- **Question Answering with LLM:** Among three user inquiries, one of the responses generated by the LLM hallucinated between two areas—criminal and corporate—due to the specific term "Probation" used in the user query. Additionally, the LLM provided generic responses instead of functioning as a specialized company bot.
- **Question Answering with LLM and Prompt Engineering:** Instances of cross-domain hallucinations have been successfully removed by the system prompt alongside the user query, allowing it to operate more like a corporate HR assistant, while still delivering basic answers without referencing corporate documentation.
- **Evaluation**
 - **Groundedness:** This is the extent to which the generated response is grounded to the context. However, the lack of context resulted in a very low score of 1 for the responses produced by both the LLM and LLM with prompt engineering.
 - **Relevance:** This is a measure of how closely the generated response relates to the user's query. In this case, the majority of the answers produced by the LLM and LLM with prompt engineering were rated between 4 and 5, meaning that they generally successfully addressed the user's query.
- **Data Preparation for RAG**
 - **Chunking:** A variety of chunking techniques were employed, including character, recursive character and semantic text splitters. The semantic method performed better because the document is chunked based on similarity between adjacent sentences, whereas the character text splitter splits words abruptly and without meaning, and the recursive character text splitter splits words fixedly but not abruptly.
 - **Embedding:** The sentences were transformed into dimensional vectors using three distinct embedding models: "thenlper/gte-large," "BAAI/bge-large-en-v1.5," and "nomic-ai/nomic-embed-text-V1.5."
 - **Vector Store:** To conduct quick similarity searches, document chunks are stored in embeddings in the "Chroma" database.
 - **Retriever:** The 'similarity' approach is used to extract relevant document chunk embeddings stored in the database by converting user queries to embeddings.
- **Question Answering using RAG:** When compared to the LLM and LLM with prompt engineering, the model performed more effectively when context was introduced and the LLM was limited to responding to user queries depending on the context. Various parameter combinations have been adjusted, and it has been noted that:
 - Because the max_tokens option is restricted, the model is unable to produce a complete response.
 - The resulting answer reflects more randomness the higher the temperature value.
 - The more tokens are repeated in the model-generated response, the lower the repeat_penalty value.
 - More diverse responses were obtained with higher top_p values, whereas less diverse responses were obtained with lower top_p values.

- While high top_k broadens the selection to a larger pool of likely tokens, enabling more diversity in the generated output, low top_k limits the model to only the few most likely next tokens, producing more concentrated and predictable content.
- **Evaluation**
 - **Groundedness:** With a rating between 4 and 5, the replies produced by the RAG model with the appropriate set of parameters behaved flawlessly as the context was revealed.
 - **Relevance:** As the answers fully addressed every important part of the user query, the final score was a perfect 5.

Recommendations

- **Integrate Contextual Retrieval for LLM Responses:** By providing the LLM with relevant context, the poor groundedness scores and generic responses can be addressed.
- **Improve Prompt Engineering with Context Retrieved:** Although domain adherence was improved via prompt engineering, it can be further improved by directly incorporating the retrieved context into the prompt. The LLM should always base its responses solely on the context given by the system prompt.
- **Improved Embedding Model:** The 'BAAI/bge-large-en-v1.5' embedding model outperformed the others, followed by 'thenlper/gte-large' and 'nomic-ai/nomic-embed-text-V1.5', when all the embedding models were compared by looking at the recovered chunks as the user query. Chunks are therefore transformed into dimensional vectors using the 'BAAI/bge-large-en-v1.5' embedding model.
- **Optimal k value:** Since the content is chunked using the semantic technique, a value of k=4 is advised in order to ensure that only relevant chunks were fetched in response to user queries.
- **Optimum parameters for RAG model:** In this instance, the RAG model is tested for many combinations, and it is advised to apply the appropriate set of parameters to address the user query in this specific circumstance, as indicated below:
 - Max tokens = 1024
 - temperature = 0
 - repeat penalty = 1.2
 - top p = 0.95
 - top k = 50
- **Deployment:** It is advised that this RAG model be used so that HR professionals can concentrate on other crucial tasks rather than responding to nearly identical questions posed by various employees. Any dynamic adjustments to the policy would also benefit greatly from it.

Appendix

Installing and Importing Necessary Libraries and Dependencies

```
# Installation for GPU llama-cpp-python
!CMAKE_ARGS="-DLLAMA_CUBLAS=on" FORCE_CMAKE=1 pip install llama-cpp-
python==0.1.85 --force-reinstall --no-cache-dir -q
# For installing the libraries & downloading models from HF hub
!pip install huggingface_hub==0.35.0 pandas==2.2.2 tiktoken==0.11.0 pymupdf==1.26.4
langchain==0.3.27 langchain-community==0.3.29 langchain_experimental==0.3.4 chromadb==1.1.0
sentence-transformers==5.1.0 numpy==2.3.3 -q
# Libraries for processing dataframes, text
import json, os
import tiktoken
import pandas as pd

# Libraries for loading data, chunking, Embedding and Vector Databases
from langchain.text_splitter import RecursiveCharacterTextSplitter, CharacterTextSplitter
from langchain_experimental.text_splitter import SemanticChunker
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.embeddings.sentence_transformer import SentenceTransformerEmbeddings
from langchain_community.vectorstores import Chroma
```

```
# Library for downloading and loading llm
from huggingface_hub import hf_hub_download
from llama_cpp import Llama
```

Question Answering using LLM

Downloading and Loading the model

```
# Loading model
model_name_or_path = 'TheBloke/Mistral-7B-Instruct-v0.2-GGUF'
model_basename = 'mistral-7b-instruct-v0.2.Q6_K.gguf' # model in gguf format
# Using hf_hub_download to download a model from the Hugging face model hub
model_path = hf_hub_download(
    repo_id = model_name_or_path, # repo id specifies the model name or path in Hugging face
    repository
    filename = model_basename # specifies name of the file to download
)
# Initialize Llama model
llm = Llama(
    model_path = model_path,
    n_ctx = 4096, # context window
    n_batch = 512, # Should be between 1 and n_ctx
    n_gpu_layers = 38 # GPU VRAM pool
)
```

Function to define model parameters for generating response

```
# Function to generate response
def response(query, max_tokens=1024, temperature=0, repeat_penalty=1.2, top_p=0.95, top_k=50):
    model_output = llm(
        prompt=query, # User query
```



```

        max_tokens = max_tokens,          # Number of tokens the response should be generated by the
model
        temperature = temperature,        # Controls the randomness of the response to be generated
        top_p = top_p,                   # Controls the diversity of the generated response by establishing
cumulative probability cut-off for token selection
        top_k = top_k,                   # Maximum number of most likely next tokens to be considered
        repeat_penalty = repeat_penalty, # Penalizes repeated tokens to improve text diversity
        stop = ['INST'],
        echo = False
    )

    return model_output['choices'][0]['text']

```

List of Questions

List of questions

```

user_queries = [
    'What are the effects on the benefits I receive if my probation is extended?',
    'There has been a demise in my family last night, and I need to attend the last rites. How should I
inform the office, and will I be granted leave?',
    'What should I do if I notice suspected harassment with my female colleague?'
]

```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```

# get the response for the user query
query_1 = user_queries[0]
response_1 = response(query=query_1)
print(response_1)

```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```

# get the response for the user query
query_2 = user_queries[1]
response_2 = response(query=query_2)
print(response_2)

```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```

# get the response for the user query
query_3 = user_queries[2]
response_3 = response(query=query_3)
print(response_3)

```

Evaluating the response generated

System message for evaluating groundedness

groundedness_rater_system_message = ""

You are tasked to rate an AI-generated answer to a question posed by the employee.

You will be presented with a question, the context used to generate the answer, and the generated answer itself.

In the input, the question begins with ###Question, the context begins with ###Context, and the AI-generated answer begins with ###Answer.

Evaluation Criteria:

Your task is to judge the extent to which the metric is followed by the answer.

Rate it 1 - The metric is not followed at all (The answer relies entirely on outside knowledge or hallucinations)

Rate it 2 - The metric is followed only to a limited extent

Rate it 3 - The metric is followed to a good extent

Rate it 4 - The metric is followed mostly

Rate it 5 - The metric is followed completely (Every single fact in the answer is explicitly backed by the context)

Metric:

Groundedness: The answer must be strictly and exclusively derived from the information presented in the `###Context`. No assumptions, extrapolations, or outside knowledge should be introduced.

Instructions:

1. First, write down the explicit steps needed to verify the truthfulness of the answer against the provided context.
2. Provide a step-by-step alignment check verifying whether the generated claims match the source context text.
3. Evaluate the extent to which the metric is followed.
4. Conclude by providing your final evaluation exactly matching the structural format requested below.

Please note: You must format the very end of your response using the exact structure below. Do not use structural headers like `###Question` or `###Context` in your final explanation.

[SCORE]: Enter the single integer rating here (1, 2, 3, 4, or 5)

[EXPLANATION]: Provide your brief explanation here.

"""

System message for evaluating relevance

relevance_rater_system_message """

You are tasked to rate the AI generated answer to the question posed by the employee.

You will be presented with a question, the context used by the AI system, and the AI-generated answer.

In the input, the question begins with `###Question`, the context begins with `###Context`, while the AI-generated answer begins with `###Answer`.

Evaluation Criteria:

Your task is to judge the extent to which the metric is followed by the answer.

Rate it 1 - The metric is not followed at all (The answer completely misses the intent of the question)

Rate it 2 - The metric is followed only to a limited extent

Rate it 3 - The metric is followed to a good extent

Rate it 4 - The metric is followed mostly

Rate it 5 - The metric is followed completely (The answer directly, fully, and comprehensively resolves the question)

Metric:

Relevance: Measures how well the answer addresses the main aspects of the question. Consider whether all and only important aspects of the question are contained in the answer.

Instructions:

1. First, write down the explicit steps needed to evaluate the answer against the question constraints.
2. Provide a step-by-step explanation of how the answer addresses the core parts of the question.
3. Next, evaluate the overall extent to which the metric is followed.

4. Conclude by providing your final evaluation exactly matching the structural format requested below.

Please note: You must format the very end of your response using the exact structure below. Do not use structural headers like '###Question' or '###Answer' in your final explanation.

[SCORE]: Enter the single integer rating here (1, 2, 3, 4, or 5)

[EXPLANATION]: Provide your brief explanation here.

"""

user template

user_message_template = """

###Question

{question}

###Context

{context}

###Answer

{answer}

"""

Function for evaluating the responses generated by AI

def eval_grounded_relevance(question, answer, context, max_tokens=1024, temperature=0, top_p=0.95, top_k=50, repeat_penalty=1.2):

"""

A function to evaluate the groundedness and relevance of the response for the user query

Key inputs required

user_query: Question posed by the user

response: Response generated by the llm

context: Context used by the llm to generate the response

"""

Groundedness evaluation

If no context or empty context is provided else device groundedness prompt

if not context or str(context).strip() == "":

Output of groundedness, if context is not presented

final_grounded_response = "Score: 1 - Evaluation bypassed because ###Context is completely empty.

No data was provided to ground the answer."

else:

Combine the user and system message to create a groundedness prompt

groundedness_prompt = f'""[INST]{groundedness_rater_system_message}

{'user': {user_message_template.format(question=question, context=context,

answer=answer)}}

[/INST]"""

Evaluating for groundedness

grounded_response = llm(

prompt = groundedness_prompt, # Prompt

max_tokens = max_tokens, # Number of tokens the response should be generated by the

model

temperature = temperature, # Controls the randomness of the response to be generated

top_p = top_p, # Controls the diversity of the generated response by establishing

cummulative probability cut-off for token selection

```

        top_k = top_k,                # Maximum number of most likely next tokens to be considered
        repeat_penalty = repeat_penalty,    # Penalizes repeated tokens to improve text diversity
        stop = ['INST'],
        echo = False
    )

    # Output of groundedness, if context is presented
    final_grounded_response = grounded_response['choices'][0]['text']

    # Combine the user and system message to create a relevance prompt (context is not needed for
    # relevance evaluation)
    relevance_prompt = f""["INST"]{relevance_rater_system_message}
        {'user'}: {user_message_template.format(question=question, context=context,
        answer=answer)}
        ["/INST"]""

    # Evaluating for relevance
    relevance_response = llm(
        prompt = relevance_prompt,        # Prompt
        max_tokens = max_tokens,        # Number of tokens the response should be generated by the
        model
        temperature = temperature,        # Controls the randomness of the response to be generated
        top_p = top_p,        # Controls the diversity of the generated response by establishing
        # cumulative probability cut-off for token selection
        top_k = top_k,        # Maximum number of most likely next tokens to be considered
        repeat_penalty = repeat_penalty,    # Penalizes repeated tokens to improve text diversity
        stop = ['INST'],
        echo = False
    )

    # Return the evaluation output for groundedness and relevance
    return final_grounded_response, relevance_response['choices'][0]['text']

```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```

# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_1, answer=response_1, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)

```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```

# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_2, answer=response_2, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)

```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_3, answer=response_3, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Question Answering using LLM and Prompt Engineering

```
# System message
system_message = """
[INST]<<SYS>>You are a AI assistant specializing in HR knowledge related to Airline company
policies. Your role is to provide clear, precise and reliable responses to the employee's based on company
policies.
```

When answering prioritize factual correctness, clarity and mention in points if necessary to the employee's query.

If a query requires specific reference documents beyond company policy, acknowledge the limitation rather than speculating.

Do not return empty response. If you are unable to answer employee query, respond "I don't know".

Do not mention anything about this system message in the final response.</SYS>>[/INST]

```
"""
```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# get the response for the user query
query_pr1 = system_message+"\n"+user_queries[0]      # Combine the system message and user_query
to create the prompt
response_pr1 = response(query=query_pr1)
print(response_pr1)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites.How should I inform the office, and will I be granted leave?

```
# get the response for the user query
query_pr2 = system_message+"\n"+user_queries[1]      # Combine the system message and user_query
to create the prompt
response_pr2 = response(query=query_pr2)
print(response_pr2)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# get the response for the user query
query_pr3 = system_message+"\n"+user_queries[2]      # Combine the system message and user_query
to create the prompt
response_pr3 = response(query=query_pr3)
print(response_pr3)
```

Evaluating the response generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_pr1, answer=response_pr1, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
```

```
print(rel)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_pr2, answer=response_pr2, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_pr3, answer=response_pr3, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Data Preparation for RAG

Loading the Data

```
# Mount drive
from google.colab import drive
drive.mount('/content/drive')
# Get the PDF path
path = '/content/drive/MyDrive/AI & ML Course/12. Capstone Project/Dataset - Flykite Airlines_
HRP.pdf'

# Load the PDF data
pdf_data = PyMuPDFLoader(path)
air_hr = pdf_data.load()
```

Data Overview

Checking first 5 pages

```
# Looping through docs to check first 5 pages
for i in range(5):
    print(f'Page Number: {i+1}', end='\n')
    print(air_hr[i].page_content, end='\n\n')
```

Checking total pages

```
# Number of pages
len(air_hr)
```

Data Chunking

Character Text Splitter

```
# Chunking the document
char_text_splitter = CharacterTextSplitter(
    separator = " , # Document separator
    chunk_size = 1000, # Maximum chunk size
    chunk_overlap = 20, # Number of tokens that overlap between the adjacent chunks
    strip_whitespace = False # Remove extra whitespaces
```

```
)

# Applying the text_splitter method on the pdf file
char_doc_chunks = pdf_data.load_and_split(char_text_splitter)

# Number of chunks the document is divided
len(char_doc_chunks)

# Checking a few document chunks from character chunking
print('Chunk 0:\n', char_doc_chunks[0].page_content, end='\n\n')
print('Chunk 1:\n', char_doc_chunks[1].page_content, end='\n\n')
print('Chunk 13:\n', char_doc_chunks[13].page_content, end='\n\n')
print('Chunk 14:\n', char_doc_chunks[14].page_content, end='\n\n')
print('Chunk 24:\n', char_doc_chunks[24].page_content, end='\n\n')
print('Chunk 25:\n', char_doc_chunks[25].page_content, end='\n\n')

Recursive Character Chunking based on Token Counts
# Chunking the document
rec_char_text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(
    encoding_name = 'cl100k_base',
    chunk_size = 100,                # Maximum chunk size
    chunk_overlap = 20              # Number of tokens that overlap between the adjacent chunks
)
```

```
# Applying the text_splitter method on the pdf file
rec_char_doc_chunks = pdf_data.load_and_split(rec_char_text_splitter)

# Number of chunks the document is divided
len(rec_char_doc_chunks)
```

```
# Checking a few document chunks from recursive character chunking
print('Chunk 0:\n', rec_char_doc_chunks[0].page_content, end='\n\n')
print('Chunk 1:\n', rec_char_doc_chunks[1].page_content, end='\n\n')
print('Chunk 13:\n', rec_char_doc_chunks[13].page_content, end='\n\n')
print('Chunk 14:\n', rec_char_doc_chunks[14].page_content, end='\n\n')
print('Chunk 24:\n', rec_char_doc_chunks[24].page_content, end='\n\n')
print('Chunk 25:\n', rec_char_doc_chunks[25].page_content, end='\n\n')
```

Semantic Chunking

```
# Initialize embedding model(used to evaluate distance between sentences)
embeddings = SentenceTransformerEmbeddings(model_name='sentence-transformers/all-MiniLM-L6-v2')

# Initialize semantic chunker
semantic_text_splitter = SemanticChunker(
    embeddings,                # Embedding model
    breakpoint_threshold_type='percentile',    # percentile method to break into chunks
    breakpoint_threshold_amount=95.0          # Splits at the 98th percentile of distance changes
)

# Creating chunks
```

```
sem_doc_chunks = pdf_data.load_and_split(semantic_text_splitter)
```

```
# Number of chunks the document is divided  
len(sem_doc_chunks)
```

```
# Checking a few document chunks from semantic chunking  
print('Chunk 0:\n', sem_doc_chunks[0].page_content, end='\n\n')  
print('Chunk 1:\n', sem_doc_chunks[1].page_content, end='\n\n')  
print('Chunk 13:\n', sem_doc_chunks[13].page_content, end='\n\n')  
print('Chunk 14:\n', sem_doc_chunks[14].page_content, end='\n\n')  
print('Chunk 24:\n', sem_doc_chunks[24].page_content, end='\n\n')  
print('Chunk 25:\n', sem_doc_chunks[25].page_content, end='\n\n')
```

Embedding

Model: 'thenlper/gte-large'

```
# Loading the embedding model
```

```
gte_embedding_model = SentenceTransformerEmbeddings(model_name='thenlper/gte-large')
```

```
# Checking for embeddings for two different document chunks
```

```
gte_embedding_1 = gte_embedding_model.embed_query(sem_doc_chunks[20].page_content)
```

```
gte_embedding_2 = gte_embedding_model.embed_query(sem_doc_chunks[21].page_content)
```

```
print(f'Dimension of the embedding model: {len(gte_embedding_1)}')
```

```
len(gte_embedding_1) == len(gte_embedding_2)
```

Model: 'BAAI/bge-large-en-v1.5'

```
# Loading the embedding model
```

```
bge_embedding_model = SentenceTransformerEmbeddings(model_name='BAAI/bge-large-en-v1.5')
```

```
# Checking for embeddings for two different document chunks
```

```
bge_embedding_1 = bge_embedding_model.embed_query(sem_doc_chunks[20].page_content)
```

```
bge_embedding_2 = bge_embedding_model.embed_query(sem_doc_chunks[21].page_content)
```

```
print(f'Dimension of the embedding model: {len(bge_embedding_1)}')
```

```
len(bge_embedding_1) == len(bge_embedding_2)
```

Model: 'nomic-ai/nomic-embed-text-V1.5'

```
# Loading the embedding model
```

```
nom_embedding_model = SentenceTransformerEmbeddings(  
    model_name='nomic-ai/nomic-embed-text-v1.5',  
    model_kwargs={'trust_remote_code': True}  
)
```

```
# Checking for embeddings for two different document chunks
```

```
nom_embedding_1 = nom_embedding_model.embed_query(sem_doc_chunks[20].page_content)
```

```
nom_embedding_2 = nom_embedding_model.embed_query(sem_doc_chunks[21].page_content)
```

```
print(f'Dimension of the embedding model: {len(nom_embedding_1)}')
```

```
len(nom_embedding_1) == len(nom_embedding_2)
```

Vector Database

```
# Create output directory
```

```
gte_out_dir = 'air_hr_man_gte'
```

```
bge_out_dir = 'air_hr_man_bge'
```



```

nom_out_dir = 'air_hr_man_nom'

# Function to embed the chunks and store them in a database
def create_store_embed(out_dir, embedding_model):
    """
    Creates embeddings for document chunks and saves them into a specific directory.
    out_dir The specific folder path for this database model.
    embedding_model: Different embedding models.
    """
    # Create the specific model output directory if it doesn't exist
    if not os.path.exists(out_dir):
        os.makedirs(out_dir)

    # Database to store document chunks
    vectorstore = Chroma.from_documents(
        sem_doc_chunks,          # Document chunks to be embedded
        embedding_model,         # Embedding function used to vectorize text chunks
        persist_directory=out_dir # Saves the database to the folder
    )

    # Load the existing stored data with the embedding model
    vectorstore = Chroma(
        persist_directory=out_dir, # Saved chunks
        embedding_function=embedding_model # Embedding model used for user query
    )
    print(vectorstore.embeddings)
    return vectorstore

# create embeddings using 'thenlper/gte-large' model and store in database
gte_vectorstore = create_store_embed(out_dir=gte_out_dir, embedding_model=gte_embedding_model)

# create embeddings using 'BAAI/bge-large-en-v1.5' model and store in database
bge_vectorstore = create_store_embed(out_dir=bge_out_dir, embedding_model=bge_embedding_model)

# create embeddings using 'nomic-ai/nomic-embed-text-V1.5' model and store in database
nom_vectorstore = create_store_embed(out_dir=nom_out_dir,
embedding_model=nom_embedding_model)

```

Retriever

```

# Retrieving the relevant document chunks
def retriever(vectorstore, k):
    """
    A function to retrieve relevant document chunks using similarity search method
    where number of relevant document chunks were varied
    vectorstore: Vector database
    k: Number of relevant document chunks to be retrieved
    """
    retriever = vectorstore.as_retriever(
        search_type = 'similarity', # Getting relevant documents using similarity search method
        search_kwargs = {'k':k}    # Getting top k relevant document chunks
    )

```

```
return retriever
```

Fetching relevant chunks when k=2 for all three embedding models

```
# Retrieving the relevant document chunks when 'thenlper/gte-large' embedding model is used
```

```
gte_retriever_2 = retriever(gte_vectorstore, 2)
```

```
# Retrieving the relevant document chunks when 'BAAI/bge-large-en-v1.5' embedding model is used
```

```
bge_retriever_2 = retriever(bge_vectorstore, 2)
```

```
# Retrieving the relevant document chunks when 'nomic-ai/nomic-embed-text-V1.5' embedding model is used
```

```
nom_retriever_2 = retriever(nom_vectorstore, 2)
```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
```

```
relevant_chunk = gte_retriever_2.get_relevant_documents(user_queries[0])
```

```
for i, doc in enumerate(relevant_chunk):
```

```
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
```

```
relevant_chunk = bge_retriever_2.get_relevant_documents(user_queries[0])
```

```
for i, doc in enumerate(relevant_chunk):
```

```
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
```

```
relevant_chunk = nom_retriever_2.get_relevant_documents(user_queries[0])
```

```
for i, doc in enumerate(relevant_chunk):
```

```
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
```

```
relevant_chunk = gte_retriever_2.get_relevant_documents(user_queries[1])
```

```
for i, doc in enumerate(relevant_chunk):
```

```
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
```

```
relevant_chunk = bge_retriever_2.get_relevant_documents(user_queries[1])
```

```
for i, doc in enumerate(relevant_chunk):
```

```
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
```

```
relevant_chunk = nom_retriever_2.get_relevant_documents(user_queries[1])
```

```
for i, doc in enumerate(relevant_chunk):
```

```
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
```

```
relevant_chunk = gte_retriever_2.get_relevant_documents(user_queries[2])
```

```

for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_2.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_2.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n")

```

Fetching relevant chunks when k=4 for all three embedding models

```

# Retrieving the relevant document chunks when 'thenlper/gte-large' embedding model is used
gte_retriever_4 = retriever(gte_vectorstore, 4)

# Retrieving the relevant document chunks when 'BAAI/bge-large-en-v1.5' embedding model is used
bge_retriever_4 = retriever(bge_vectorstore, 4)

# Retrieving the relevant document chunks when 'nomic-ai/nomic-embed-text-V1.5' embedding model is used
nom_retriever_4 = retriever(nom_vectorstore, 4)

```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```

# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_4.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_4.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_4.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n")

```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```

# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_4.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_4.get_relevant_documents(user_queries[1])

```

```

for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_4.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```

# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_4.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_4.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_4.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

```

Fetching relevant chunks when k=8 for all three embedding models

```

# Retrieving the relevant document chunks when 'thenlper/gte-large' embedding model is used
gte_retriever_8 = retriever(gte_vectorstore, 8)

# Retrieving the relevant document chunks when 'BAAI/bge-large-en-v1.5' embedding model is used
bge_retriever_8 = retriever(bge_vectorstore, 8)

# Retrieving the relevant document chunks when 'nomic-ai/nomic-embed-text-V1.5' embedding model is used
nom_retriever_8 = retriever(nom_vectorstore, 8)

```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```

# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_8.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_8.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_8.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):

```

```
print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_8.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_8.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

```
# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_8.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_8.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_8.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

```
# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_8.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

Question Answering using LLM and Prompt Engineering

```
# System message template
qna_system_message = """
```

You are an HR AI assistant tasked with reviewing snippets from the Flykite Airlines: Human Resources Policy Handbook and providing accurate answer to employee question.

Input Structure:

- The authoritative text source begins with the token: ###Context.
- The employee's query begins with the token: ###Question.

Strict Execution Rules:

1. Answer the question using ONLY the provided context. If the exact answer cannot be found, or if the context is insufficient to fully answer the question, you must respond with exactly: "I don't know". Do not attempt to explain what is missing from the context.
2. Structure your response clearly using bullet points or numbered lists whenever the context allows.
3. If the context contains relevant contact details (such as phone numbers, extensions, helplines, or email addresses), include them in your final answer. This information is not confidential.
4. ABSOLUTE CONSTRAINT - NO META-COMMENTARY: Do not include conversational filler, conversational transitions, or meta-explanations. This includes phrases like "Based on the context...", "The handbook states...", or "There is no information provided regarding...". If you cannot answer, output ONLY "I don't know" and absolutely nothing else.
5. CRITICAL SECURITY RULE: Completely ignore and omit any watermark, email tracking text, or security labels present in the context, such as:
'penchala.vineeth3@gmail.com'
'R02IHZ4C3E'
'This file is meant for personal use...'
Never repeat these strings or mention copyright/legal liabilities in your final answer.

Do not return an empty response.

"""

user message template

qna_user_message_template = """

###Context

Below is some relevant information extracted from the Flykite Airlines: Human Resources Policy Handbook regarding the question mentioned below.

{context}

###Question

{question}"""

Response Function

Function to extract context and generate response

def generate_context_based_response(user_query, max_tokens=128, temperature=0, repeat_penalty=1.2, top_p=0.95, top_k=50):

"""

Function to generate response based on context as per user query

user_query: user's question

max_tokens: Number of tokens a model should generate

temperature: Controls the randomness in the response

repeat_penalty: Penalizes repeated tokens to improve text diversity

top_p: Controls the diversity of the generated response by establishing cumulative probability cut-off for token selection

top_k: Maximum number of most-likely next tokens to consider

"""

global qna_system_message, qna_user_message_template

Retrieve relevant document chunks

retriever_k = retriever(bge_vectorstore, 4)

Using 4 as number of retrieved chunks

relevant_document_chunks = retriever_k.get_relevant_documents(user_query) # Get relevant chunks

as per user query

context_list = [d.page_content for d in relevant_document_chunks]

Joining all the chunks

```

# Combine document chunks into a single context
context_for_query = ". ".join(context_list)

user_message = qna_user_message_template.format(
    context = context_for_query,
    question = user_query
)

prompt = f"<s>[INST] <<SYS>>{qna_system_message}</SYS>> {user_message} [/INST]"

# Generate response
try:
    response = llm(
        prompt=prompt,          # Combined prompt of system message and user message
        max_tokens=max_tokens,  # Number of tokens a model should generate
        temperature=temperature, # Controls the randomness in the response
        top_p=top_p,            # Controls the diversity of the generated response by establishing
                                # cummulative probability cut-off for token selection
        top_k=top_k,            # Maximum number of most-likely next tokens to consider
        repeat_penalty=repeat_penalty, # Penalizes repeated tokens to improve text diversity
        echo=False
    )

    # Extract and print models response
    response = response['choices'][0]['text']
except Exception as e:
    response = f"Sorry, I encountered the following error: \n {e}"
return response, context_for_query

```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```

# Get the answer for user query
query_1 = user_queries[0]
answer_1, context_1 = generate_context_based_response(query_1)
print(answer_1)

```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```

# Get the answer for user query
query_2 = user_queries[1]
answer_2, context_2 = generate_context_based_response(query_2)
print(answer_2)

```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```

# Get the answer for user query
query_3 = user_queries[2]
answer_3, context_3 = generate_context_based_response(query_3)
print(answer_3)

```

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_1, answer=answer_1, context=context_1)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_2, answer=answer_2, context=context_2)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_3, answer=answer_3, context=context_3)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Fine-tuning (Case: 1): max_tokens=256, temperature=1, top_p=0.5, top_k=90, repeat_penalty=0

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Get the answer for user query
answer_f1_1, context_f1_1 = generate_context_based_response(query_1, max_tokens=256,
temperature=1, top_p=0.5, top_k=90, repeat_penalty=0)
print(answer_f1_1)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Get the answer for user query
answer_f1_2, context_f1_2 = generate_context_based_response(query_2, max_tokens=256,
temperature=1, top_p=0.5, top_k=90, repeat_penalty=0)
print(answer_f1_2)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Get the answer for user query
```



```
answer_f1_3, context_f1_3 = generate_context_based_response(query_3, max_tokens=256,
temperature=1, top_p=0.5, top_k=90, repeat_penalty=0)
print(answer_f1_3)
```

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_1, answer=answer_f1_1, context=context_f1_1)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_2, answer=answer_f1_2, context=context_f1_2)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_3, answer=answer_f1_3, context=context_f1_3)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Fine-tuning (Case: 2): max_tokens=512, temperature=0.8, top_p=0.8, top_k=70, repeat_penalty=0.5

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Get the answer for user query
answer_f2_1, context_f2_1 = generate_context_based_response(query_1, max_tokens=512,
temperature=0.8, top_p=0.8, top_k=70, repeat_penalty=0.5)
print(answer_f2_1)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Get the answer for user query
answer_f2_2, context_f2_2 = generate_context_based_response(query_2, max_tokens=512,
temperature=0.8, top_p=0.8, top_k=70, repeat_penalty=0.5)
print(answer_f2_2)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Get the answer for user query
answer_f2_3, context_f2_3 = generate_context_based_response(query_3, max_tokens=512,
temperature=0.8, top_p=0.8, top_k=70, repeat_penalty=0.5)
print(answer_f2_3)
```

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_1, answer=answer_f2_1, context=context_f2_1)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_2, answer=answer_f2_2, context=context_f2_2)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_3, answer=answer_f2_3, context=context_f2_3)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Fine-tuning (Case: 3): max_tokens=512, temperature=0.5, top_p=0.8, top_k=80, repeat_penalty=0.8

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Get the answer for user query
answer_f3_1, context_f3_1 = generate_context_based_response(query_1, max_tokens=512,
temperature=0.5, top_p=0.8, top_k=80, repeat_penalty=0.8)
print(answer_f3_1)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Get the answer for user query
```

```
answer_f3_2, context_f3_2 = generate_context_based_response(query_2, max_tokens=512,
temperature=0.5, top_p=0.8, top_k=80, repeat_penalty=0.8)
print(answer_f3_2)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Get the answer for user query
answer_f3_3, context_f3_3 = generate_context_based_response(query_3, max_tokens=512,
temperature=0.5, top_p=0.8, top_k=80, repeat_penalty=0.8)
print(answer_f3_3)
```

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_1, answer=answer_f3_1, context=context_f3_1)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_2, answer=answer_f3_2, context=context_f3_2)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_3, answer=answer_f3_3, context=context_f3_3)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Fine-tuning (Case: 4): max_tokens=1024, temperature=0.2, top_p=0.95, top_k=90, repeat_penalty=1

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Get the answer for user query
answer_f4_1, context_f4_1 = generate_context_based_response(query_1, max_tokens=1024,
temperature=0.2, top_p=0.95, top_k=80, repeat_penalty=1)
print(answer_f4_1)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Get the answer for user query
answer_f4_2, context_f4_2 = generate_context_based_response(query_2, max_tokens=1024,
temperature=0.2, top_p=0.95, top_k=80, repeat_penalty=1)
print(answer_f4_2)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Get the answer for user query
answer_f4_3, context_f4_3 = generate_context_based_response(query_3, max_tokens=1024,
temperature=0.2, top_p=0.95, top_k=80, repeat_penalty=1)
print(answer_f4_3)
```

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_1, answer=answer_f4_1, context=context_f4_1)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_2, answer=answer_f4_2, context=context_f4_2)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_3, answer=answer_f4_3, context=context_f4_3)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Fine-tuning (Case: 5): max_tokens=1024, temperature=0, top_p=0.95, top_k=50, repeat_penalty=1.2

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Get the answer for user query
```

```
answer_f5_1, context_f5_1 = generate_context_based_response(user_query=query_1, max_tokens=1024,
temperature=0, top_p=0.95, top_k=50, repeat_penalty=1.2)
print(answer_f5_1)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Get the answer for user query
answer_f5_2, context_f5_2 = generate_context_based_response(user_query=query_2, max_tokens=1024,
temperature=0, top_p=0.95, top_k=50, repeat_penalty=1.2)
print(answer_f5_2)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Get the answer for user query
answer_f5_3, context_f5_3 = generate_context_based_response(user_query=query_3, max_tokens=1024,
temperature=0, top_p=0.95, top_k=50, repeat_penalty=1.2)
print(answer_f5_3)
```

Evaluation

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_1, answer=answer_f5_1, context=context_f5_1)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_2, answer=answer_f5_2, context=context_f5_2)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_3, answer=answer_f5_3, context=context_f5_3)
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)
```